PROVEIT RESEARCH CONTINUATION

Exact Harmonic-Tail Reconstruction and Variational Gamma Identities

Gap-four determinants, polynomial telescopers,
higher resonances, polylogarithms, and Stieltjes coefficients

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Research draft prepared with ChatGPT

11 October 2026

Repository baseline. The analysis uses the exact ProveIt revision `446dca9bd65c20e409d1d256 1d4ffad408103e2a`, including the manuscript and all six archives then present in `docs/incoming`. The accompanying source manifest records paths, Git blob identifiers and SHA-256 hashes.

Scope of the claims. The report proves the gap-four subfamily of the optimal moment-prefix conjecture and derives additional exact identities. The arbitrary unequal-order conjecture and the short Gaussian S_6 and current S_8 evaluations remain open. Formal ranks refer to explicitly chosen coefficient spaces. Numerical diagnostics are reported separately from proofs. Historical novelty beyond the inspected sources is not asserted.

Abstract

This continuation develops four related methods for exact identities. First, a fourth-order Bernoulli operator proves that the first q centered harmonic-tail moments have full formal rank for every even pair $p = q - 4 \geq 2$. The remaining boundary determinant is controlled by a strict ratio-monotonicity argument, not by extrapolating finite ranks. Second, every finite late-moment combination for arbitrary positive even orders admits an explicit polynomial-tail telescoping certificate. Its negative even Laurent coefficients are exactly the zeta coordinates; cancellation of the positive zeta coordinates gives a rational evaluation. Third, the Euler variational derivative yields an exact three-variable equation for the complete ψ, ψ', ψ'' sector of the inverse-Gamma hierarchy. We solve the square and cube sectors in closed form and give rational coefficient formulas at every order. Fourth, the next rank-two Gamma-weight resonance is evaluated with all cutoff and spectral counterterms retained, producing polylogarithmic generating functions and convergent zeta evaluations. A final Mellin calculation relates higher spectral derivatives to explicitly subtracted logarithmic spatial integrals and Stieltjes coefficients. The report specifies the inherited results, the new proof scopes, integration recommendations, and further research targets.

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1 Research scope, conventions, and dependencies

The repository has a substantial body of overlapping reports. A useful continuation must first fix its comparison point. We use the commit on the title page, whose last change added six new research bundles. The canonical manuscript already incorporates many older results; the six new bundles contain additional theorems that must also be credited before identifying an open question. The baseline and its mathematical status are recorded in [1, 2] and in the individual reports [3, 4, 5, 6, 7, 8].

1.1 What is continued here

The optimal-prefix question concerns the centered moments of two ordinary zeta tails. The incoming reports prove the diagonal case, the adjacent even-order case $p = q - 2$, and the families with smaller order $2, 4$, or 6; they leave a general unequal-order determinant unresolved [3, 6, 8]. Section 2 proves the entire next gap $q - p = 4$. Thus its first pair outside those earlier small-order families is $(p, q) = (8, 12)$. Section 3 then turns late-coordinate relations into explicit telescopers for all even pairs, whether or not their optimal-prefix determinant is known to be nonzero.

The inverse-Gamma normal form and the full digamma–trigamma sector are inherited from [6, 8]. The explicit request in [8, research questions 4–5] asks for the next mixed sector, including arbitrary powers of the first two derivative variables. Section 4 gives an exact recurrence for the entire sector with largest derivative index at most two. It also proves closed formulas for all terms containing exactly two or exactly three copies of ψ''. This is a coefficient theorem in a differential polynomial algebra; its integration consequences retain every endpoint term.

The rank-two higher-resonance problem is stated explicitly in [5, Section 9]. Section 6 treats that target, including the logarithmic square in the hard cutoff and its difference from the spectral Laurent constant. The logarithmically weighted finite-part problem in [6] is treated in Section 7. These are analytic continuations with specified counterterms, so their normalization is part of each theorem.

1.2 Ordinary tails, strict sums, and spectral derivatives

For $s > 1$ and $n \geq 0$, put

$$H_n^{(s)} = \sum_{k=1}^{n} \frac{1}{k^s}, \qquad A_s(n) = \zeta(s) - H_n^{(s)} = \zeta(s, n+1). \tag{1.1}$$

For integer $s \geq 2$ this gives

$$A_s(n) = \frac{(-1)^s}{(s-1)!} \psi^{(s-1)}(n+1).$$

The even orders used in the tail sections therefore have positive polygamma normalization. These are standard Hurwitz identities [10, Sections 5.15 and 25.11].

A multiple zeta value uses strictly decreasing indices:

$$\zeta(s_1, \dots, s_d) = \sum_{n_1 > \cdots > n_d \geq 1} n_1^{-s_1} \cdots n_d^{-s_d}$$

where the series converges, with meromorphic continuation when needed. A prime on a function of a spectral variable denotes differentiation in that variable. In particular it never denotes differentiation of a formula known only at integer index values.

We use generalized Stieltjes constants with the convention

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{j\ge 0}\frac{(-1)^j}{j!}\gamma_j(a)u^j,\qquad a>0, \tag{1.2}$$

and $\gamma_j=\gamma_j(1)$. Thus $\gamma_0(a)=-\psi(a)$.

1.3 Two precise meanings of finite part

For a function with a finite Laurent-logarithmic singular expansion at zero, its original-coordinate integral finite part is

$$\mathrm{FP}\int_0^1 f(x)\,\mathrm{d}x=\mathrm{CT}_{\varepsilon\downarrow 0}\int_\varepsilon^1 f(x)\,\mathrm{d}x. \tag{1.3}$$

The symbol CT selects the constant multiplying $\varepsilon^0(\log\varepsilon)^0$. The coordinate is x itself. For a polynomial H in digamma derivatives this convention implies

$$\mathrm{FP}\int_0^1 \frac{d}{dx}H(\psi(x),\psi'(x),\dots)\,\mathrm{d}x=H(\psi(1),\psi'(1),\dots)-[x^0]H(\psi(x),\psi'(x),\dots). \tag{1.4}$$

This follows directly by taking the constant in the endpoint difference. A change of coordinate must be evaluated separately.

The centered tail moment is a different, explicitly specified object:

$$D_{p,q}(s)=\sum_{n\ge 0}A_p(n)A_q(n)(n+\tfrac12)^{-s},\qquad \operatorname{Re}s>3-p-q. \tag{1.5}$$

We continue it meromorphically and write $D_r=D_{p,q}(-2r)$. For positive even p,q, the centered asymptotic expansion has only even powers; the possible poles are at odd integers, so all these values are regular. Equivalently, subtract from $(n+\frac12)^{2r}A_p(n)A_q(n)$ its polynomial part in $n+\frac12$. The resulting summand is $O(n^{-2})$, and its ordinary sum equals D_r because $\zeta(-2j,\frac12)=0$ for $j\ge 0$. Section 3 proves this relation with the finite-cutoff constants made explicit.

1.4 Status and reproducibility

The proofs below establish their all-order claims algebraically or analytically. Exact rational verification files check the matrix conjugacies, polynomial certificates, normal forms, and displayed specializations. Independent numerical computations compare different analytic representations. They are diagnostics, not interval proofs, unless a check is explicitly described as rational arithmetic.

No assertion of arithmetic independence follows from a formal-rank statement. In particular, the symbols used as late zeta coordinates are treated as formal basis labels even though their numerical values have known multiplicative relations. Likewise, the new recurrence for Gamma coefficients does not prove that the corresponding mixed moments form a minimal numerical constant basis.

2 The optimal moment prefix for every even-tail gap of four

The incoming *Differential Polygamma and Lerch Identities* and *Polygamma Tail Identities* reports prove the optimal consecutive moment prefix for adjacent even orders, $q-p=2$, and explicitly leave the next gap $q-p=4$ open. The incoming *Bernoulli, Tornheim, and Digamma Identities* report proves all pairs with $p=2,4,6$ and gives a residual determinant of size $p/2$ for arbitrary even

pairs. We prove the entire gap-four family. The first case outside those already proved families is $(p, q) = (8, 12)$ [6, 8, 3].

The Differential report already reduces a fixed gap to a residual determinant of size $(q - p)/2$, whose columns satisfy a positive recurrence. We rederive that reduction with its normalization visible, then prove the missing strict separation of column ratios. This shows that the two-column determinant cannot vanish. These are statements about rational coefficient matrices; no arithmetic independence of zeta values is asserted. The fixed-gap input is [6, `ar:gap-det`, `ar:gap-factor`].

2.1 The inherited moment coordinates and their conventions

For even integers $p = 2a$, $q = 2b$, with $1 \le a \le b$, write

$$A_p(n) = \zeta(p, n + 1) = \zeta(p) - H_n^{(p)}, \qquad x_n = n + \tfrac{1}{2},$$

and let D_r denote the continued value at $s = -2r$ of

$$D_{p,q}(s) = \sum_{n \ge 0} A_p(n) A_q(n) x_n^{-s}.$$

The source results prove regularity at these even arguments and the inverse Bernoulli transforms

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j + 1} B_{2r-2j}(1/2) C_j, \tag{2.1}$$

$$C_j = \sum_{r=0}^{j} \binom{2j + 1}{2r} 2^{-2(j-r)} D_r. \tag{2.2}$$

For $j \ge b$ their exact late coordinates are

$$C_j = \sum_{k=0}^{b-1} a_{jk} \zeta(2k), \qquad \zeta(0) = -\tfrac{1}{2}. \tag{2.3}$$

With $d = 2j + 1$ and $e = d - p - q + 2k + 1$, the inherited entries are zero for $e < 0$, and otherwise

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d - q + 1} + \frac{\binom{d-p+1}{q-2k}}{d - p + 1} \right\}. \tag{2.4}$$

Each binomial summand is omitted unless its lower index lies between 2 and its upper index. In particular, lower index zero is omitted. The Bernoulli convention is $B_1 = -1/2$.

Set

$$M_{a,b} = (a_{b+r,k})_{0 \le r, k < b}. \tag{2.5}$$

In the formal coordinate order

$$C_0, \dots, C_{b-1}, \quad \zeta(0), \zeta(2), \dots, \zeta(2b - 2),$$

the determinant of the complete first-q moment matrix is

$$\frac{\det M_{a,b}}{(4b - 1)!!}. \tag{2.6}$$

Indeed the first b rows of the C coordinate matrix form an identity block, the last b rows have the block $M_{a,b}$, and the forward transform has diagonal $1/(2r + 1)$. These facts and (2.1)–(2.4) are source inputs; the nonvanishing argument that follows is the new result.

2.2 A complementary determinant of size half the gap

Put $m = b - a \geq 1$ and $h(t) = \coth(t/2)$. For an even polynomial $Q(t) = \sum_{n=1}^{b} q_n t^{2n}$ define

$$\mathcal{L}_m Q(t) = (Q^{(2m)}(t) - Q^{(2m)}(0))h(t) + (Q(t)h(t))^{(2m)}. \tag{2.7}$$

This is an odd formal Taylor series: the subtraction cancels the possible pole in the first product, and Qh is an odd Taylor series. Let $K_{m,b}$ map $(q_1, \ldots, q_b)$ to the coefficients of $t, t^3, \ldots, t^{2b-1}$ in (2.7).

Lemma 2.1 (Exact operator conjugacy)**.** *If $M_{a,b}^{\mathrm{rev}}$ reverses the columns of $M_{a,b}$, then*

$$K_{m,b} = \operatorname{diag}_{0 \leq r < b}\left(\frac{2}{(2r+1)!}\right) M_{a,b}^{\mathrm{rev}} \operatorname{diag}_{1 \leq n \leq b}((2n)!). \tag{2.8}$$

Consequently

$$\det K_{m,b} = (-1)^{b(b-1)/2} 4^b b! \det M_{a,b}. \tag{2.9}$$

Proof. Write

$$h(t) = \sum_{k \geq 0} h_k t^{2k-1}, \qquad h_0 = 2, \quad h_k = \frac{2B_{2k}}{(2k)!} \quad (k \geq 1).$$

For use in all subsequent formulas define

$$A_r = \begin{cases} 0, & 1 \leq r \leq m, \\ (2r)^{\underline{2m}}, & r \geq m+1, \end{cases} \qquad B_N = (2N-1)^{\underline{2m}}. \tag{2.10}$$

The underline denotes a falling factorial. Row r of $K_{m,b}$ corresponds to $N = r + m + 1$. Its entry in column n is

$$h_{N-n}(A_n + B_N) \quad (n \leq N), \tag{2.11}$$

and is zero for $n > N$. The exceptional value $A_m = 0$ comes from the subtraction $Q^{(2m)}(0)$.

To compare this with the source, put $j = b + r$ and $k = b - n$ in (2.4). The Bernoulli index becomes $2(N - n)$, while the two upper and lower binomial indices are

$$(2r + 2,\, 2n - 2m), \qquad (2r + 2m + 2,\, 2n).$$

Multiplying by $2(2n)!/(2r+1)!$ gives exactly (2.11). The first summand is omitted for $n \leq m$, precisely as required by (2.10). This proves (2.8). The product of its positive diagonal factors is

$$2^b \prod_{n=1}^{b} \frac{(2n)!}{(2n-1)!} = 4^b b!,$$

and column reversal has sign $(-1)^{b(b-1)/2}$. □

Put

$$\beta_k = \frac{2|B_{2k}|}{(2k)!} = \frac{4\zeta(2k)}{(2\pi)^{2k}} > 0 \quad (k \geq 1), \tag{2.12}$$

and, for $N \geq m + 1$, put

$$d_N = 2\left\{(2N)^{\underline{2m}} + (2N-1)^{\underline{2m}}\right\} > 0. \tag{2.13}$$

Define an array of rational numbers $C_{N,j}^{[m]}$, with $N \geq 1$ and $1 \leq j \leq m$, by

$$C_{r,j}^{[m]} = \delta_{rj} \qquad (1 \leq r \leq m), \tag{2.14}$$

$$C_{N,j}^{[m]} = \frac{1}{d_N} \sum_{r=1}^{N-1} \beta_{N-r}(A_r + B_N) C_{r,j}^{[m]} \qquad (N \geq m+1). \tag{2.15}$$

Every entry $C_{N,j}^{[m]}$ with $N \geq m+1$ is strictly positive. The superscript distinguishes this array from the inherited moment coordinates C_j.

Theorem 2.2 (Inherited fixed-gap boundary reduction)**.** *For $1 \leq a < b$ and $m = b - a$,*

$$\det M_{a,b} = \frac{(-1)^{b(b-1)/2}}{4^b b!} \left(\prod_{N=m+1}^{b+m} d_N \right) \det(C_{b+i,j}^{[m]})_{1 \leq i,j \leq m}. \tag{2.16}$$

Thus the remaining determinant has size $m = (q-p)/2$, independently of the larger order q.

Proof. The sign relation $h_k = (-1)^{k-1}\beta_k$ for $k \geq 1$ shows that

$$Q_j(t) = \sum_{n \geq 1} (-1)^{n-j} C_{n,j}^{[m]} t^{2n} \qquad (1 \leq j \leq m) \tag{2.17}$$

formally satisfy $\mathcal{L}_m Q_j = 0$. Conversely, the coefficient at $t^{2N-2m-1}$ determines the next unknown coefficient uniquely: its nonzero leading factor is d_N. Hence these are the unique formal null solutions with their specified first m coefficients. This argument requires no convergence of the formal series.

Replace the first m columns of $K_{m,b}$ by the corresponding linear combinations specified by the truncations of (2.17) through $n = b$. The column transformation has determinant one, since those truncations have initial block equal to the identity. Their first $b - m$ rows vanish. Their bottom m rows form the block $-TQ_{\text{future}}$, where T is lower triangular with diagonal $d_{b+1}, \ldots, d_{b+m}$, and

$$(Q_{\text{future}})_{ij} = (-1)^{b+i-j} C_{b+i,j}^{[m]}.$$

This follows by writing the omitted terms in each exact formal coefficient equation; row $b+i$ omits only coefficients $b+1, \ldots, b+i$.

Move the first m columns to the end. The other diagonal block is triangular with diagonal $d_{m+1}, \ldots, d_b$. Therefore

$$\det K_{m,b} = (-1)^{m(b-m)+m} \left(\prod_{N=m+1}^{b+m} d_N \right) \det Q_{\text{future}}.$$

Factoring the row and column signs in Q_{future} gives $\det Q_{\text{future}} = (-1)^{mb} \det(C_{b+i,j}^{[m]})$. The combined exponent $2mb - m(m-1)$ is even. Apply (2.9) to obtain (2.16). □

2.3 Strict separation of the recurrence columns

The classical Bernoulli Fourier formula, DLMF 24.8.1 at zero, gives the second equality in (2.12) [10]. In particular, with $x_\ell = (2\pi\ell)^{-2}$,

$$\beta_k = 4 \sum_{\ell \geq 1} x_\ell^k \qquad (k \geq 1).$$

Strict Cauchy–Schwarz implies

$$\beta_{k+1}^2 < \beta_k \beta_{k+2} \qquad (k \geq 1), \tag{2.18}$$

since the positive support contains distinct points. Thus $\rho_k = \beta_{k+1}/\beta_k$ is strictly increasing. This familiar moment argument is used solely to certify nonzero exact determinants.

Lemma 2.3 (Ratio separation for every gap)**.** *Let $m \geq 2$. For any two initial columns $1 \leq i < j \leq m$, the sequence*

$$R_N = \frac{C_{N,i}^{[m]}}{C_{N,j}^{[m]}} \qquad (N \geq m+1) \tag{2.19}$$

is strictly increasing. Consequently every two-by-two minor of the positive part of this array, with both its row indices and its initial-column indices increasing, is strictly negative.

Proof. All relevant entries are positive. Since $A_1 = \cdots = A_m = 0$, divide the recurrence by the second column's initial forcing $B_N \beta_{N-j}/d_N$. Set

$$f_N = \frac{\beta_{N-i}}{\beta_{N-j}}, \qquad w_{N,r} = \frac{\beta_{N-r}}{\beta_{N-j}} \left(1 + \frac{A_r}{B_N}\right). \tag{2.20}$$

The exact ratio equation is

$$R_N = \frac{f_N + \displaystyle\sum_{r=m+1}^{N-1} w_{N,r} C_{r,j}^{[m]} R_r}{1 + \displaystyle\sum_{r=m+1}^{N-1} w_{N,r} C_{r,j}^{[m]}}. \tag{2.21}$$

By (2.18), f_N strictly increases: it is a fixed-length product of consecutive ρ_k. For every fixed $r \geq m+1 > j$, the quotient β_{N-r}/β_{N-j} is the reciprocal of a fixed-length product of consecutive ρ_k, so it strictly decreases as N increases from $r+1$. Moreover $A_r > 0$ and B_N strictly increases. Thus $w_{N,r}$ strictly decreases with N for each such fixed r.

At $N = m+1$ the sums are empty, giving $R_{m+1} = f_{m+1}$. For the induction step assume the ratios through R_{N-1} are strictly increasing. Multiply (2.21) by its denominator, subtract R_{N-1} times that denominator, and subtract the preceding row's corresponding zero identity. The term $r = N-1$ vanishes. The result is

$$\begin{aligned} &\left(1 + \sum_{r=m+1}^{N-1} w_{N,r} C_{r,j}^{[m]}\right)(R_N - R_{N-1}) \\ &\quad = f_N - f_{N-1} + \sum_{r=m+1}^{N-2} (w_{N,r} - w_{N-1,r}) C_{r,j}^{[m]} (R_r - R_{N-1}) > 0. \end{aligned} \tag{2.22}$$

The first term is strictly positive. Each term in the sum is nonnegative, because both differences there are negative. This also handles the first induction step, when the sum is empty. The assertion about minors follows by factoring out the positive entries in their second column. □

2.4 The gap-four determinant and the optimal prefix

Specialize to $m = 2$. Write

$$u_N = C_{N,1}^{[2]}, \qquad v_N = C_{N,2}^{[2]}.$$

For explicit calculation their common recurrence is

$$\begin{aligned}
&(u_1,u_2)=(1,0), \qquad (v_1,v_2)=(0,1),\\
&y_N=\frac{\displaystyle\sum_{r=1}^{N-1}\beta_{N-r}(A_r+B_N)y_r}{16(N-1)^2(2N-1)(2N-3)} \qquad (N\ge 3),\\
&A_1=A_2=0, \qquad A_r=(2r)(2r-1)(2r-2)(2r-3) \quad (r\ge 3),\\
&B_N=(2N-1)(2N-2)(2N-3)(2N-4).
\end{aligned} \tag{2.23}$$

Define the strictly positive rational number

$$\Delta_b=u_{b+2}v_{b+1}-u_{b+1}v_{b+2}>0. \tag{2.24}$$

Its sign follows immediately from Lemma 2.3.

Theorem 2.4 (Every pair $q=p+4$)**.** *For every integer $b\ge 3$, let $p=2b-4$, $q=2b$. Then*

$$\det M_{b-2,b}=(-1)^{b(b-1)/2+1}\frac{4^b((b+1)!)^2(2b+3)!!(2b+1)!!}{3b!}\,\Delta_b\neq 0. \tag{2.25}$$

The moments $D_0,\dots,D_{q-1}$ therefore have the full formal rank q. Every later centered moment is an explicit rational linear combination of these first q moments. The determinant of their full coordinate matrix is (2.25) *divided by* $(4b-1)!!$.

Proof. In (2.16) the endpoint determinant is $-\Delta_b$. Here

$$d_N=16(N-1)^2(2N-1)(2N-3),$$

so direct multiplication gives

$$\prod_{N=3}^{b+2}d_N=\frac{16^b((b+1)!)^2(2b+3)!!(2b+1)!!}{3}.$$

This proves (2.25). Its scalar factor is nonzero and $\Delta_b>0$, establishing nonvanishing. The full rank assertion follows from (2.6). To reconstruct a later moment, use (2.2) to obtain $C_0,\dots,C_{2b-1}$ from the first $2b$ moments, invert $M_{b-2,b}$ to recover all the even-zeta coordinates, and then use (2.3) and (2.1). Every step is finite rational arithmetic. □

2.5 A compact ordinary identity for the first new pair

For $(p,q)=(8,12)$ the determinant and first inverse row are

$$\begin{aligned}
\det M_{4,6}&=\frac{263589503}{309657600},\\
(M_{4,6}^{-1})_{0,*}&=\frac{8}{790768509}(469839,1269227,-11819495,56443363,83672540,12834702).
\end{aligned} \tag{2.26}$$

Thus the displayed integer row applied to $C_6,\dots,C_{11}$ has value $-790768509/16$.

Define the odd polynomial

$$\begin{aligned}
K(N)=N^{13}(&469839+1269227N^2-11819495N^4+56443363N^6\\
&+83672540N^8+12834702N^{10})
\end{aligned} \tag{2.27}$$

and its centered difference

$$W(x) = K(x + \tfrac{1}{2}) - K(x - \tfrac{1}{2}). \tag{2.28}$$

The resulting ordinary, absolutely convergent identity is

$$\sum_{n=0}^{\infty} \left\{ W(x_n)A_8(n)A_{12}(n) - \frac{295198146}{77}x_n^4 - \frac{10254604347}{154}x_n^2 - \frac{59115956909}{385} \right\} = -\frac{790768509}{16}. \tag{2.29}$$

The summand is $O(x_n^{-2})$. To see this directly, the centered Hurwitz expansion gives the first three terms of $A_8(n)A_{12}(n)$; multiplication by the degree-22 polynomial W gives precisely the three displayed nondecaying terms. The inverse Bernoulli transform identifies its continued sum with the integer combination in (2.26); the removed even powers have continued sum zero because $\zeta(-2r, 1/2) = 0$ for $r \geq 0$. The accompanying finite rational telescoping certificate in Section 3.1 supplies a second proof of this individual ordinary identity.

In polygamma notation the tail product is

$$A_8(n)A_{12}(n) = \frac{\psi^{(7)}(n+1)\psi^{(11)}(n+1)}{7!\,11!}.$$

The compact difference definition (2.28) avoids listing twelve large coefficients while specifying the weight exactly.

2.6 Verification, source update, and remaining limits

The supplied exact script independently constructs the source matrix from its binomial formula, constructs the operator matrix by symbolic differentiation on monomials, and constructs the two recurrence solutions. Its 323 rational checks verify conjugacy for $3 \leq b \leq 12$, the determinant formula for $3 \leq b \leq 18$, formal null-series coefficients through $N = 24$, and the finite sign relations used in the proof. These computations are diagnostics for finite formulas; the proof above establishes the infinite family. A second standalone replay checks 45 complementary determinants for all $1 \leq a < b \leq 10$ and 140 pairwise ratio minors at gaps $m = 2, 3, 4, 5$. Thus the two tail scripts contain 508 exact checks in total; none extrapolates its finite range into a proof.

The optimal-prefix source conjecture can now record $q - p = 4$ as proved, alongside the inherited equal-order, adjacent-order, and $p = 2, 4, 6$ families. The fixed-gap reduction (2.16) is the Differential report's Proposition `ar:gap-det`, rederived here; it retains that attribution. The new input is Lemma 2.3, which settles its two-dimensional boundary determinant. No new false inherited theorem was established in this targeted audit.

For $m \geq 3$, pairwise ratio separation does not by itself imply nonvanishing of the m-square endpoint determinant. In fact its sign need not remain fixed as b changes. The next precise target is the $m = 3$ determinant, corresponding to $q - p = 6$, with the already proved overlapping pairs retained. Further useful targets are a generating function for the two universal solutions in (2.23), and an exact recurrence or arithmetic normalization for Δ_b itself. A general higher-minor sign claim would require an additional proof; it is not a consequence of the positivity used here.

3 A universal finite telescoper for late harmonic-tail moments

This section resolves the systematic telescoping question in [6]. The coordinate formulas used in its application are inherited from [6, 3]; the finite certificate and its limiting calculation are given here for arbitrary positive even orders.

Let $2 \le p \le q$ be even integers, $b = q/2$, and

$$A_r(N) = \zeta(r) - H_N^{(r)} = \sum_{k>N} k^{-r}.$$

Choose finitely many rational coefficients λ_j, $b \le j \le J$, and put

$$K(X) = \sum_{j=b}^{J} \lambda_j X^{2j+1}, \qquad P(X) = K(X + \tfrac{1}{2}) - K(X - \tfrac{1}{2}).$$

In particular P is even. Define Faulhaber polynomials by their values on nonnegative integers:

$$U(N) = \sum_{k=1}^{N} \frac{K(k)}{k^q}, \qquad V(N) = \sum_{k=1}^{N} \frac{K(k)}{k^p}.$$

The conditions $j \ge b$ ensure that the summands are polynomials in k with positive odd powers. Thus $U, V \in \mathbb{Q}[N]$ and $U(0) = V(0) = 0$.

Define the finite even Laurent polynomial

$$H(N) = \frac{K(N)}{N^{p+q}} + \frac{U(N-1)}{N^p} + \frac{V(N-1)}{N^q} = H_+(N) + \sum_{k=1}^{b-1} \frac{h_k}{N^{2k}}, \qquad h_0 = H_+(0), \tag{3.1}$$

where H_+ contains the nonnegative powers. Let $S(X)$ be the nonnegative-power part of the centered asymptotic expansion of $P(X)A_p(X - 1/2)A_q(X - 1/2)$. This is an explicitly computable even polynomial: one uses only finitely many terms of

$$A_r(X - \tfrac{1}{2}) \sim \frac{X^{1-r}}{r-1} + \sum_{\ell \ge 1} \frac{B_{2\ell}(1/2)(r)_{2\ell-1}}{(2\ell)!} X^{1-r-2\ell}. \tag{3.2}$$

Finally let $R \in \mathbb{Q}[N]$ be the unique polynomial with

$$R(0) = 0, \qquad R(N) - R(N-1) = H_+(N) - S(N - \tfrac{1}{2}).$$

Theorem 3.1 (A finite certificate and its limiting value)**.** *With these definitions,*

$$F(N) = K(N)A_p(N)A_q(N) + U(N)A_p(N) + V(N)A_q(N) + R(N) \tag{3.3}$$

satisfies the exact finite identity

$$F(N) - F(N-1) = P(N - \tfrac{1}{2})A_p(N-1)A_q(N-1) - S(N - \tfrac{1}{2}) - \sum_{k=1}^{b-1} \frac{h_k}{N^{2k}}. \tag{3.4}$$

Moreover $F(0) = 0$ and $\lim_{N\to\infty} F(N) = -h_0/2$. Consequently the ordinary absolutely convergent sum has the exact value

$$\boxed{\sum_{n=0}^{\infty} \Big\{ P(n + \tfrac{1}{2})A_p(n)A_q(n) - S(n + \tfrac{1}{2}) \Big\} = -\frac{h_0}{2} + \sum_{k=1}^{b-1} h_k \zeta(2k).} \tag{3.5}$$

If $h_1 = \cdots = h_{b-1} = 0$, the certificate consists entirely of polynomial multiples of the product of the two tails, the two individual tails, and a polynomial; its value is the rational number $-h_0/2$.

Proof. For a positive odd integer m, write

$$\sum_{k=1}^{N} k^m = E_m(N) + \frac{1}{2}N^m, \qquad E_m(-N) = E_m(N), \quad E_m(0) = 0.$$

This follows immediately by expanding $\{B_{m+1}(N+1) - B_{m+1}\}/(m+1)$: apart from its half-power term, all surviving powers are positive even powers. Hence there are even polynomials U_e, V_e, both vanishing at zero, such that

$$U(N) = U_e(N) + \frac{K(N)}{2N^q}, \qquad V(N) = V_e(N) + \frac{K(N)}{2N^p}.$$

Their difference laws imply $U(N-1) = U_e(N) - K(N)/(2N^q)$, and similarly for V. Therefore

$$H(N) = \frac{U_e(N)}{N^p} + \frac{V_e(N)}{N^q}. \tag{3.6}$$

This proves its evenness. Since U_e, V_e have no constant term, the most negative possible exponent is $2-q$, which gives the stated range $1 \le k \le b-1$.

Set $a = A_p(N)$ and $b' = A_q(N)$ and use $A_p(N-1) = a + N^{-p}$ and $A_q(N-1) = b' + N^{-q}$. Expanding $F(N) - F(N-1)$ and using

$$\Delta K = P(N - \tfrac{1}{2}), \quad \Delta U = K(N)N^{-q}, \quad \Delta V = K(N)N^{-p}$$

proves (3.4) as a rational-function identity in N, a, b'. No identity among zeta values enters this step. The centered expansion (3.2) shows that its first two terms together are $O(N^{-2})$; the residual even reciprocal powers are summable. Thus $F(N)$ has a finite limit.

It remains to compute that limit without analytic continuation. The standard uncentered asymptotic expansion has the parity form

$$A_r(N) \sim O_r(N) - \frac{1}{2}N^{-r}, \qquad O_r(-N) = -O_r(N),$$

where O_r is an asymptotic Laurent series containing only odd negative powers. Combining all three expressions gives the exact formal asymptotic identity

$$KA_pA_q + UA_p + VA_q \sim KO_pO_q + U_eO_p + V_eO_q - \frac{K}{4}N^{-p-q} - \frac{U_e}{2}N^{-p} - \frac{V_e}{2}N^{-q}.$$

The first four terms have odd powers only. The surviving constant is

$$-\frac{1}{2}\left([N^p]U_e + [N^q]V_e\right) = -\frac{h_0}{2}$$

by (3.6). The polynomial R has zero constant. All computations use finitely many asymptotic terms, and the remainder can be taken $o(1)$ after multiplication by the fixed polynomials. Since the limit exists, positive powers in this expansion cancel and the limit equals the displayed constant. Summing (3.4) and using $F(0) = 0$ proves (3.5). □

Exact agreement with the inherited late-block matrix. For $d = 2j+1$ and $e = d-p-q+2k+1$, the coefficient of N^{-2k} in the contribution of N^d to H is zero for $e < 0$, and otherwise

$$a_{jk} = B_e \left\{ \frac{\binom{d-q+1}{p-2k}}{d-q+1} + \frac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}.$$

Each summand is omitted unless its lower binomial index is between two and its upper index. Indeed U_e, V_e contain positive even powers only; the forbidden lower-index-zero term is absent because their constant terms are zero. Thus $h_k = \sum_j \lambda_j a_{jk}$, including $k = 0$. This is exactly the normalization in Differential equations `ar:entries`, `ar:block`, and Bernoulli equation `eq:general-source-entries` [6, 3]. The theorem supplies the all-order finite certificate requested in Differential Section 09, "Systematic rational telescoping for tail products." It proves the certificate for every even pair and every late-block combination, without presupposing a nonsingular optimal prefix matrix.

Degree bounds. If $\lambda_J \neq 0$, then

$$\deg K = 2J+1, \quad \deg P = 2J, \quad \deg U = 2J+2-q, \quad \deg V = 2J+2-p.$$

Put $\delta = 2J+2-p-q$. If $\delta \geq 0$, then $\deg H_+, \deg S \leq \delta$ and $\deg R \leq \delta+1$. If $\delta < 0$, $H_+ = S = R = 0$. For the optimal prefix $J = q-1$ these bounds become $\deg S \leq q-p$ and $\deg R \leq q-p+1$.

3.1 A finite certificate for the (8,12) identity

For the polynomial K in (2.27), set $P = W$ as in (2.28). The exact identity is (2.29), with $A_8(n)A_{12}(n) = \psi^{(7)}(n+1)\psi^{(11)}(n+1)/(7!\,11!)$.

Here is a compact certificate retaining every rational coefficient. Put $T = N(N+1)$ and define

$$\begin{aligned} U(N) &= \frac{T}{8}\big(8556468T^5 + 32712160T^4 - 38171739T^3 \\ &\qquad + 24232272T^2 - 9577682T + 1879356\big), \end{aligned} \tag{3.7}$$

$$V(N) = \frac{T^5}{8}\big(6417351T^3 - 3525928T^2 - 1698454T + 1879356\big),$$

$$\begin{aligned} R(N) &= -\frac{N}{6160}\big(2417202210N^4 - 5764920315N^3 \\ &\qquad + 42395905270N^2 - 129829345800N + 260578219219\big). \end{aligned} \tag{3.8}$$

The corresponding Laurent polynomial actually has no negative powers:

$$H(N) = \frac{3}{8}\left(4991273N^4 + 112406360N^2 + 263589503\right).$$

Hence $h_0 = 790768509/8$ and $h_k = 0$ for $1 \leq k \leq 5$. Substitution of (2.27) and (3.7)–(3.8) into (3.3) gives a finite identity whose difference is exactly the summand of (2.29), and whose limit is the right side. All its steps are rational-function identities followed by the parity calculation in the theorem. The accompanying symbolic program checks the identity in independent variables for $A_8(N), A_{12}(N)$, and separately checks the limiting constant from finite Laurent expansions.

4 A variational equation for all Gamma coefficients of derivative order at most two

This section answers the next-sector question in [8, research questions 4–5]. The result is independent of the tail determinant: it is an exact identity in a free differential polynomial algebra. After substitution of digamma derivatives, it provides coefficients in identities among Gamma, polygamma, and height-one zeta integrals.

4.1 The inherited normal form and the new finite-jet principle

Let

$$\mathscr{R} = \mathbb{Q}[u_0, u_1, u_2, \ldots], \qquad Du_r = u_{r+1}, \qquad \mathrm{wt}(u_r) = r + 1.$$

A monomial is *functional* if it is a power of u_0, including 1, or if its largest positive derivative index occurs at least twice. Write $\mathscr{F}$ for their rational linear span. The classical functional-monomial decomposition [9, Proposition 4.10], used in [6, 8], is

$$P = \mathcal{N}(P) + D\mathcal{H}(P), \qquad \mathcal{N}(P) \in \mathscr{F}, \qquad [1]\mathcal{H}(P) = 0. \tag{4.1}$$

It is unique. For completeness, if a monomial M has unique largest index $r \geq 1$, replace u_r by u_{r-1} to form T, and let a be the multiplicity of u_{r-1} in T. Then

$$M = \frac{1}{a}DT + \left(M - \frac{1}{a}DT\right).$$

Every remaining monomial has smaller largest derivative index. This proves existence by termination. If H is nonconstant and has largest derivative index r, the nonzero term $u_{r+1}\partial H/\partial u_r$ in DH has unique largest index; it cannot belong to $\mathscr{F}$. This proves uniqueness. Both maps preserve factor degree, while D raises weight by one. The quotient is a vector-space quotient: $\mathcal{N}$ is linear but is not multiplicative. For example, $\mathcal{N}(u_1) = 0$ while $\mathcal{N}(u_1^2) = u_1^2$.

Define the Euler variational derivative

$$\mathscr{E}(P) = \sum_{r\geq 0}(-D)^r \frac{\partial P}{\partial u_r}. \tag{4.2}$$

The sum is finite for each polynomial. The commutator identity $\partial_{u_r} D = D\partial_{u_r} + \partial_{u_{r-1}}$, with the last term zero for $r = 0$, proves by cancellation that $\mathscr{E}(DH) = 0$.

For $d \geq 0$ let σ_d set all u_r, $r > d$, equal to zero, and let

$$\delta_d = \sum_{r=0}^{d-1} u_{r+1}\partial_{u_r}$$

act on $\mathbb{Q}[u_0, \ldots, u_d]$.

Lemma 4.1 (Finite-jet variational principle)**.** *For every polynomial P,*

$$\sigma_d\mathscr{E}(P) = \sum_{r=0}^{d}(-\delta_d)^r\partial_{u_r}(\sigma_d\mathcal{N}(P)). \tag{4.3}$$

Thus the variational derivative determines a closed equation for a truncated normal form even though normal reduction itself must precede truncation.

Proof. Equation (4.1) and $\mathscr{E}D = 0$ permit replacing P by $\mathcal{N}(P)$ on the left. Any normal monomial removed by σ_d has its highest index greater than d at least twice. Any partial derivative in (4.2) removes at most one such factor. Every subsequent application of D preserves at least one factor of index greater than d. Its specialization is therefore zero. For the retained terms, the ideal generated by $u_{d+1}, u_{d+2}, \dots$ is D-stable, so specialization of D^r is δ_d^r. These observations prove the identity term by term. □

4.2 The inverse-Gamma hierarchy

Set

$$\mathcal{A}_0 = 1, \qquad \mathcal{A}_{m+1} = u_0\mathcal{A}_m - D\mathcal{A}_m. \tag{4.4}$$

The Bell exponential is

$$\mathcal{A}(t) = \sum_{m\geq 0} \mathcal{A}_m \frac{t^m}{m!} = \exp\left(\sum_{r\geq 0} \frac{(-1)^r u_r t^{r+1}}{(r+1)!}\right). \tag{4.5}$$

Under $u_r = \psi^{(r)}(x)$, these become

$$\mathcal{A}_m(x) = (-1)^m \Gamma(x) \frac{d^m}{dx^m} \frac{1}{\Gamma(x)}, \qquad \mathcal{A}(t,x) = \frac{\Gamma(x)}{\Gamma(x-t)}.$$

All operations below are formal in t; no interchange with an infinite integral is needed for the coefficient theorem.

Translation $u_0 \mapsto u_0 + c$ commutes with D and preserves $\mathscr{F}$, hence commutes with $\mathcal{N}$. It follows that $\mathcal{N}(\mathcal{A}(t))$ has an Appell factor $e^{u_0 t}$. There is also the centered identity

$$\mathcal{N}(\mathcal{A}(t)) = \mathcal{N} \exp\left(\sum_{j\geq 0} \frac{u_{2j} t^{2j+1}}{2^{2j}(2j+1)!}\right). \tag{4.6}$$

Indeed the exponential on the right before normal reduction is $e^{tD/2}\mathcal{A}(t)$, and every coefficient of $(e^{tD/2} - 1)\mathcal{A}(t)$ is a total derivative. Consequently every retained monomial has factor degree congruent to its weight modulo two.

There are uniquely determined formal series $f_b(y)$ such that

$$\sigma_2\mathcal{N}(\mathcal{A}(t)) = e^{u_0 t} f(y,z), \qquad f(y,z) = \sum_{b\geq 0} z^b f_b(y), \qquad y = \frac{u_1 t^2}{2}, \quad z = u_2 t^3. \tag{4.7}$$

Each f_b is even in y. The functional-monomial convention excludes one copy of u_2, so $f_1 = 0$.

Theorem 4.2 (Complete three-variable Gamma-sector equation)**.** *Put $L = 2y + (z/2)\partial_y$. Then*

$$\boxed{f - \frac{1}{2}Lf_y + L^2 f_z = e^{z/24} \int_0^1 \cosh(yv) e^{zv^2/8}\, \mathrm{d}v.} \tag{4.8}$$

The equation, the normal-form conditions $f_1 = 0$ and absence of a linear y term, and the normalization $f_0(0) = 1$ uniquely determine all the functions in (4.7). More explicitly, let

$$R_n(y) = \frac{1}{n! 24^n} \int_0^1 (1+3v^2)^n \cosh(yv)\, \mathrm{d}v. \tag{4.9}$$

Then

$$f_0(y) = \frac{1}{2}\left(\cosh y + \frac{\sinh y}{y} - y\,\mathrm{Shi}(y)\right), \qquad f_1(y) = 0, \quad (4.10)$$

$$4(n+1)y^2 f_{n+1}(y) = R_n(y) - (n+1)f_n(y) - (2n-1)yf_n'(y) - \frac{n-2}{4}f_{n-1}''(y), \qquad n \geq 1. \quad (4.11)$$

Here $\mathrm{Shi}(y) = \int_0^y \sinh v\,\mathrm{d}v/v$. *Every apparent singularity at* $y = 0$ *in these formulas is removable.*

Proof. Differentiating (4.5) with respect to a variable u_r and applying the Euler operator gives

$$\mathscr{E}\mathcal{A}(t) = \sum_{r\geq 0}\frac{t^{r+1}}{(r+1)!}D^r\mathcal{A}(t) = \int_0^t e^{sD}\mathcal{A}(t)\,\mathrm{d}s. \quad (4.12)$$

After setting $u_r = 0$ for $r \geq 3$, a formal shift replaces u_0 by $u_0 + su_1 + s^2u_2/2$ and u_1 by $u_1 + su_2$. The exponent of $e^{sD}\mathcal{A}(t)$ becomes

$$tu_0 + (st - t^2/2)u_1 + (ts^2/2 - t^2s/2 + t^3/6)u_2.$$

Substitute $s = t(1+v)/2$. The exponent is $tu_0 + yv + z/24 + zv^2/8$. Integrating over $-1 \leq v \leq 1$ and using its even part yields the right side of (4.8), multiplied by te^{tu_0}.

On $e^{tu_0}g(y,z)$ the restricted derivation is $e^{tu_0}t^{-1}Lg$. Therefore Lemma 4.1 gives

$$\sigma_2\mathscr{E}\mathcal{N}\mathcal{A}(t) = te^{tu_0}\{f - \tfrac{1}{2}Lf_y + L^2f_z\}.$$

This proves (4.8). Expanding

$$L^2 = 4y^2 + z + 2zy\partial_y + \frac{z^2}{4}\partial_y^2$$

and comparing z^n gives (4.11). At z^0 the condition $f_1 = 0$ gives $f_0 - yf_0' = \sinh y/y$. Its even solution with constant one is (4.10); the only homogeneous freedom is a multiple of y, which the normal form excludes.

For $n \geq 1$, (4.11) determines f_{n+1} uniquely because multiplication by y^2 is injective on formal series. Divisibility of its numerator by y^2 follows from the existence of the normal form already proved in (4.1). This is not an assumption about arbitrary initial functions. The coefficient theorem below also proves that each f_b is entire in y. □

4.3 Rational formulas for every coefficient

For even $k \geq 0$, write

$$r_b(k) = k![y^k]f_b(y).$$

Then $r_0(k) = -1/((k-1)(k+1))$ and $r_1(k) = 0$. Taking coefficients in (4.11) gives the explicit rational recurrence

$$r_{n+1}(k) = \frac{1}{4(n+1)(k+1)(k+2)}\Bigg\{\frac{1}{n!24^n}\sum_{j=0}^{n}\binom{n}{j}\frac{3^j}{k+2j+3} - (n+1+(2n-1)(k+2))r_n(k+2) - \frac{n-2}{4}r_{n-1}(k+4)\Bigg\}. \quad (4.13)$$

This computes a whole coefficient family with rational operations, without expanding all partitions at its total weight.

Theorem 4.3 (Polynomial numerators and fixed odd denominators)**.** *For every $b \geq 2$ there is a rational polynomial $p_b(X)$ of degree exactly $b-2$ such that*

$$\boxed{r_b(k) = \frac{p_b(k)}{\prod_{j=0}^{b}(k+2j+1)}, \qquad [X^{b-2}]p_b(X) = \frac{1}{4b!6^{b-1}}.} \tag{4.14}$$

These formulas hold for all even $k \geq 0$. In particular every $f_b(y)$ is entire. If $a, k \geq 0$ and $b \geq 2$, then

$$[u_0^a u_1^k u_2^b]\mathcal{N}(\mathcal{A}_{a+2k+3b}) = \begin{cases} \dfrac{(a+2k+3b)!}{a!2^k k!}\dfrac{p_b(k)}{\prod_{j=0}^{b}(k+2j+1)}, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases} \tag{4.15}$$

A coefficient with one copy of u_2 is zero.

Proof. The case $b = 2$ follows from (4.13): $p_2 = 1/48$. For a polynomial recurrence, set

$$\Delta_n(X) = \prod_{h=0}^{n}(X+2h+3), \qquad S_n(X) = \frac{1}{n!24^n}\sum_{j=0}^{n}\binom{n}{j}3^j\frac{\Delta_n(X)}{X+2j+3}.$$

Each quotient in this expression is a polynomial. For $n \geq 2$, (4.13) becomes

$$\begin{aligned} p_{n+1}(X) = \frac{1}{4(n+1)(X+2)}\Big\{ & S_n(X) - (n+1+(2n-1)(X+2))p_n(X+2) \\ & - \frac{n-2}{4}(X+3)p_{n-1}(X+4)\Big\}, \end{aligned} \tag{4.16}$$

where the last term is zero for $n = 2$. The numerator vanishes at $X = -2$: after division by $\Delta_n(-2) \neq 0$, that assertion is exactly the constant-term identity in (4.11), whose left side contains y^2. Thus (4.16) is a polynomial recurrence, not merely a rational-function definition.

The degree of S_n is n, and its leading coefficient is $4^n/(n!24^n) = 1/(n!6^n)$. The two subtracted terms have degree at most $n-1$ and $n-2$. Division by $4(n+1)(X+2)$ proves that p_{n+1} has degree $n-1$ and the stated leading coefficient. Induction proves (4.14). For fixed b, $r_b(k) = O(k^{-3})$ as $k \to \infty$, and the factorial in the series for f_b proves entire convergence. Finally substitute $y = u_1 t^2/2$, $z = u_2 t^3$ in (4.7) and expand $e^{u_0 t}$. The factor-degree parity already proved gives the zero odd coefficients. □

The first polynomial numerators are

$$\begin{aligned} p_2(X) &= \frac{1}{48}, & p_3(X) &= \frac{2X+5}{1728}, \\ p_4(X) &= \frac{X^2+7X+9}{20736}, & p_5(X) &= \frac{2X^3+27X^2+100X+90}{1244160}, \\ p_6(X) &= \frac{2X^4+44X^3+316X^2+832X+603}{44789760}. \end{aligned}$$

The theorem supplies every subsequent numerator by the same finite recurrence. It does not require that all its displayed possible poles be actual poles; some factors can cancel at higher b.

4.4 Closed square and cube sectors

Corollary 4.4 (All mixed tetragamma-square and cube coefficients)**.** *The two first new entire functions are*

$$f_2(y) = \frac{(y^2+3)\sinh y - 3y\cosh y}{48y^5} = \frac{1}{384}\int_0^1 (1-v^2)^2\cosh(yv)\,\mathrm{d}v, \tag{4.17}$$

$$f_3(y) = \frac{(2y^4+60y^2+135)\sinh y - 15y(y^2+9)\cosh y}{1728y^7}. \tag{4.18}$$

In particular, for all $a, s \geq 0$,

$$[u_0^a u_1^{2s} u_2^2]\mathcal{N}(\mathcal{A}_{a+4s+6}) = \frac{(a+4s+6)!}{48a!2^{2s}(2s)!(2s+1)(2s+3)(2s+5)}, \tag{4.19}$$

$$[u_0^a u_1^{2s} u_2^3]\mathcal{N}(\mathcal{A}_{a+4s+9}) = \frac{(4s+5)(a+4s+9)!}{1728a!2^{2s}(2s)!(2s+1)(2s+3)(2s+5)(2s+7)}. \tag{4.20}$$

The removable values are $f_2(0) = 1/720$ and $f_3(0) = 1/36288$.

Proof. The numerator formulas for p_2, p_3 give their power series directly. For the square sector use

$$\int_0^1 v^k(1-v^2)^2\,\mathrm{d}v = \frac{8}{(k+1)(k+3)(k+5)}.$$

For the cube sector use

$$f_3(y) = \frac{2y\partial_y + 5}{82944}\int_0^1 (1-v^2)^3\cosh(yv)\,\mathrm{d}v.$$

Repeated elementary integration gives (4.17) and (4.18). Coefficient extraction in (4.15) proves the final formulas. □

For example, the coefficients of u_2^3 in $\mathcal{N}(\mathcal{A}_9)$, of $u_1^2u_2^2$ in $\mathcal{N}(\mathcal{A}_{10})$, and of u_2^4 in $\mathcal{N}(\mathcal{A}_{12})$ are respectively $10, 90$, and 220. At weight 14, the coefficient of $u_1^4u_2^2$ is 15015. These are consequences of the all-order formulas, not the evidence from which their validity is inferred.

4.5 What the sector theorem does and does not evaluate

The specialization σ_2 is performed after normal reduction. Substituting $u_r = \psi^{(r)}(x)$ into a full moment identity does not make the omitted derivatives vanish. For instance

$$\mathcal{N}(\mathcal{A}_8) = u_0^8 - 70u_0^4u_1^2 + 28u_0^2u_2^2 - 7u_1^4 - u_3^2.$$

The last term remains in the digamma identity. The new theorem solves the coefficient problem for a complete sector and supplies a finite-jet variational method for higher sectors; it does not claim that the sector alone evaluates an arbitrary pure digamma moment. The next section states the exact transfer with its full boundary functional.

5 Exact Gamma moments and their endpoint terms

The coefficient theorem has an immediate analytic use, but it is essential to keep the complete normal form and its primitive. We give the inherited scalar bridge with its normalization, then an explicit ninth-weight identity containing the first cube term. The bridge itself is already proved in [8, Section 4].

5.1 Height-one Laurent coefficients as ordinary integrals

Define rational zeta polynomials c_j and height-one Laurent coefficients λ_d by

$$\frac{1}{\Gamma(1+u)} = \sum_{j\ge 0} c_j u^j, \qquad \lambda_d = [u^1]\zeta(1+u,\{1\}^{d-1}), \qquad d \ge 1. \tag{5.1}$$

The coefficient on the right is a Laurent coefficient: for $d > 1$ the pole at zero has order d. In particular $c_0 = 1$, $c_1 = \gamma$, and $\lambda_1 = -\gamma_1$. For $v > 0$ put

$$L(v) = -\log(1-e^{-v}), \qquad V_d(v) = L(v)^d - \mathbf{1}_{(0,1)}(v)(-\log v)^d.$$

Then the normalization can be stated without an unevaluated finite part:

$$\boxed{\lambda_d = c_{d+1} + \frac{1}{d!}\int_0^\infty \frac{V_d(v)}{v}\,\mathrm{d}v.} \tag{5.2}$$

The integral is ordinary and absolutely convergent. At zero its integrand is $O((1+|\log v|)^{d-1})$; at infinity it decays exponentially up to the factor $1/v$.

Here is a direct derivation. With $H_{n-1}(\{1\}^{d-1})$ denoting the strict unit-index multiple harmonic sum, its generating function is

$$\sum_{n\ge 1} H_{n-1}(\{1\}^{d-1})e^{-nv} = -\frac{1}{d!}\frac{d}{dv}L(v)^d.$$

This follows by differentiating the power series for $L(v)^d$, or inductively from strict summation. Mellin integration and integration by parts, initially for $\operatorname{Re} u > 0$, give

$$\begin{aligned}\zeta(1+u,\{1\}^{d-1}) &= \frac{1}{d!\Gamma(u)}\int_0^\infty v^{u-1}L(v)^d\,\mathrm{d}v \\ &= \frac{1}{\Gamma(1+u)u^d} + \frac{u}{d!\Gamma(1+u)}\int_0^\infty v^{u-1}V_d(v)\,\mathrm{d}v.\end{aligned} \tag{5.3}$$

The last integral is holomorphic for u in a neighborhood of zero. Extracting $[u^1]$ proves (5.2) and fixes its sign.

5.2 The scalar inverse-Gamma evaluation

Let $\mathcal{A}_m$ be the differential polynomials of (4.4). For $m \ge 2$ set

$$\mathcal{L}_m = \mathrm{FP}\int_0^1 \mathcal{A}_m(x)\,\mathrm{d}x, \qquad K_{2r} = \frac{(-1)^r\pi^{2r}}{2r+1}, \quad K_{2r+1} = 0.$$

The inherited all-order scalar formula is

$$\boxed{\mathcal{L}_m = K_m - m!\sum_{d=1}^{m-1}(-1)^{m-1-d}c_{m-1-d}\lambda_d.} \tag{5.4}$$

We recall why no coordinate constant is suppressed. Put

$$\mathcal{C}_m(x) = m![t^m]\frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)}.$$

For $m \geq 2$, $\mathcal{C}_m$ extends continuously to $[0,1]$. Indeed the residue of the displayed quotient at $x = 0$ is exactly $-t$, with no higher powers of t. Euler's beta integral applied to $F(x,t) = \Gamma(1+t)\Gamma(x)/\Gamma(x+t)$ yields

$$\int_0^1 \{F(x,t) - 1 + t(\psi(x)+\gamma)\}\,\mathrm{d}x = t\int_0^\infty \frac{(1-e^{-v})^t - 1 - \mathbf{1}_{(0,1)}(v)(v^t-1)}{v}\,\mathrm{d}v. \tag{5.5}$$

For clarity, the subtraction follows by integrating in x first when $\operatorname{Re} t > 0$. The resulting expression is

$$t\int_0^\infty \left\{\frac{(1-e^{-v})^t - 1}{v} + \frac{e^{-v}}{1-e^{-v}}\right\}\mathrm{d}v - 1.$$

At lower cutoff δ, the second integral is $-\log(1-e^{-\delta})$, while $\int_\delta^1 (v^t-1)\,\mathrm{d}v/v = (1-\delta^t)/t + \log\delta$. Their limits prove (5.5). Its subtracted integrand also justifies holomorphic continuation and coefficient extraction near $t = 0$.

Replacing t by $-t$ and using (5.2) now gives

$$\int_0^1 \mathcal{C}_m(x)\,\mathrm{d}x = -m\int_0^\infty \frac{V_{m-1}(v)}{v}\,\mathrm{d}v = m!(c_m - \lambda_{m-1}), \qquad m \geq 2.$$

The constant and linear coefficientwise finite parts of the normalized quotient are 1 and γ, since $\mathcal{C}_1 = \psi + \gamma$ and $\mathrm{FP}\int_0^1 \psi(x)\,\mathrm{d}x = 0$. Thus its complete coefficientwise integral is

$$\frac{1}{\Gamma(1+t)} - \sum_{d\geq 1} \lambda_d t^{d+1}.$$

Multiplication by $1/\Gamma(1-t)$ and the reflection identity $1/(\Gamma(1+t)\Gamma(1-t)) = \sin(\pi t)/(\pi t)$ prove (5.4). All uses of an infinite generating series here are coefficientwise identities justified by the displayed local subtractions.

5.3 The complete differential identity and a cube-sector moment

Write $P_m = \mathcal{N}(\mathcal{A}_m)$ and $H_m = \mathcal{H}(\mathcal{A}_m)$. With

$$\Delta H = H(\psi(1), \psi'(1), \ldots) - [x^0]H(\psi(x), \psi'(x), \ldots), \tag{5.6}$$

the exact transfer is

$$\boxed{\mathrm{FP}\int_0^1 P_m(\psi, \psi', \ldots)\,\mathrm{d}x = K_m - m!\sum_{d=1}^{m-1}(-1)^{m-1-d}c_{m-1-d}\lambda_d - \Delta H_m.} \tag{5.7}$$

This follows from $\mathcal{A}_m = P_m + DH_m$ and (1.4). The endpoint is finite algebra: the standard local series [10, Section 5.7] is

$$\psi(x) = -x^{-1} - \gamma + \sum_{j\geq 1}(-1)^{j+1}\zeta(j+1)x^j.$$

For a monomial $M = \prod u_r^{a_r}$ of weight $w = \sum(r+1)a_r$, multiply its substitutions by x^w and take the coefficient of x^w. Only the first $w+1$ coefficients of each normalized series $x^{r+1}\psi^{(r)}(x)$ can contribute. This gives a terminating algorithm in $\mathbb{Q}[\gamma, \zeta(2), \ldots, \zeta(w)]$.

For example, the full ninth-weight normal polynomial is

$$P_9 = u_0^9 - 126u_0^5u_1^2 + 84u_0^3u_2^2 - 63u_0u_1^4 - 9u_0u_3^2 + 10u_2^3. \tag{5.8}$$

The coefficient 10 is the first cube term from Corollary 4.4. For a fully explicit endpoint write $Z_j = \zeta(j)$ and define

$$\begin{aligned} B_9 = & -252\gamma^6 Z_2 + 1260\gamma^5 Z_3 + \gamma^4(3150Z_2^2 - 5670Z_4) \\ & + \gamma^3(-17640Z_2Z_3 + 20328Z_5) \\ & + \gamma^2(-7560Z_2^3 + 40572Z_2Z_4 + 16884Z_3^2 - 57456Z_6) \\ & + \gamma(28476Z_2^2Z_3 - 70560Z_2Z_5 - 54432Z_3Z_4 + 106416Z_7) \\ & + 1932Z_2^4 - 22050Z_2^2Z_4 - 18648Z_2Z_3^2 + 63036Z_2Z_6 \\ & + 47376Z_3Z_5 + 21942Z_4^2 - 99624Z_8. \end{aligned} \tag{5.9}$$

Then $B_9 = \Delta H_9$, and the concrete identity is

$$\begin{aligned} \mathrm{FP}\int_0^1 \{\psi^9 - 126\psi^5(\psi')^2 + 84\psi^3(\psi'')^2 - 63\psi(\psi')^4 - 9\psi(\psi''')^2 + 10(\psi'')^3\}\,\mathrm{d}x \\ = -9!\sum_{d=1}^{8}(-1)^{8-d}c_{8-d}\lambda_d - B_9. \end{aligned} \tag{5.10}$$

All functions inside the integral have argument x. The formula specifies an evaluated combination of mixed moments; it is not a claim that each term separately reduces to ordinary zeta values. If desired, every λ_d on its right can be replaced by the convergent integral (5.2).

For comparison, the next full normal polynomial is

$$\begin{aligned} P_{10} = & u_0^{10} - 210u_0^6u_1^2 + 210u_0^4u_2^2 - 315u_0^2u_1^4 - 45u_0^2u_3^2 \\ & + 100u_0u_2^3 + 90u_1^2u_2^2 + u_4^2. \end{aligned} \tag{5.11}$$

The companion exact data file gives P_m, H_m through weight 14, both endpoint polynomials at weights $9, 10$, and the polynomial numerators p_b of Theorem 4.3 through $b = 10$. The verification program checks the complete differential equality $\mathcal{A}_m = P_m + DH_m$ through weight 24, not only the retained three-variable sector.

5.4 Anchored primitives and higher derivative sectors

For any real $x, x_0 > 0$, the same identity gives the ordinary anchored primitive relation

$$\int_{x_0}^{x} P_m(\psi(t), \psi'(t), \ldots)\,\mathrm{d}t = \int_{x_0}^{x} \mathcal{A}_m(t)\,\mathrm{d}t - H_m(\psi(x), \psi'(x), \ldots) + H_m(\psi(x_0), \psi'(x_0), \ldots). \tag{5.12}$$

It states exactly how the primitive changes under normal reduction; it does not postulate an elementary antiderivative for $\mathcal{A}_m$. Section 6 gives a separate polylogarithmic family whose repeated Euler primitives and convergent endpoints are explicit. For additional differential sectors, Lemma 4.1 applies without change with $d = 3, 4, \ldots$. The resulting equations involve more variables but retain all variational endpoint information.

6 The next two-particle resonance: all overlap terms and its cutoff constant

The incoming *Determinantal Harmonic Resonance* report [5, Section 9, first proposed target] proves the first resonance and explicitly proposes the two-particle next resonance as its initial higher-resonance target. We resolve that target here. The input Gauss summation and the determinant identity are classical; the contribution relative to that report is a normally convergent subtraction at the next resonance, including the product overlap, an explicit hard-cutoff constant, and its exact comparison with the spectral Laurent constant. No historical priority claim is intended.

Let $\Re a, \Re c > 0$, and put

$$b_0 = a + c + 1, \qquad w_n(u) = \frac{\Gamma(n+a)\Gamma(n+c)}{\Gamma(n+b_0+u)\Gamma(n+1)}, \qquad n \geq 0.$$

The two-particle series

$$Z(u) = \sum_{0 \leq m < n} (n-m)^2 w_m(u) w_n(u)$$

initially converges for $\Re u > 1$. The next resonance is $u = 0$. Use the abbreviations

$$t = ac + a + c, \qquad A = \frac{1}{ac}, \qquad h = \frac{t}{ac} = 1 + \frac{1}{a} + \frac{1}{c},$$
$$L(u) = \frac{\Gamma(a+1)\Gamma(c+1)\Gamma(1+u)}{\Gamma(a+1+u)\Gamma(c+1+u)}, \qquad A(u) = \frac{L(u)}{ac},$$
$$B = \psi(1) - \psi(a+1) - \psi(c+1), \qquad T = \psi'(1) - \psi'(a+1) - \psi'(c+1).$$

Thus $L(0) = 1$, $L'(0) = B$, and $L''(0) = B^2 + T$. For an independent anchor $\Re\alpha > 0$, define

$$\kappa_\alpha(u) = -t + \left(\alpha - a - c - \tfrac{1}{2}\right) u - \tfrac{1}{2}u^2.$$

Powers of $n + \alpha$ use the analytic logarithm on the right half-plane, real on the positive real axis. For $x > 0$, write $x^{-u} = \exp(-u \log x)$ with the real logarithm.

Theorem 6.1 (Normally convergent higher-resonance subtraction)**.** *For $\Re u > -1$, the following three sums converge locally normally, jointly in a, c, u, α on the stated right-half-plane domains:*

$$A(u) = \sum_{n \geq 0} w_n(u), \tag{6.1}$$

$$B_\alpha(u) = \sum_{n \geq 0} \left[n w_n(u) - (n+\alpha)^{-1-u} \right], \tag{6.2}$$

$$D_\alpha(u) = \sum_{n \geq 0} \left[n^2 w_n(u) - (n+\alpha)^{-u} - \kappa_\alpha(u)(n+\alpha)^{-1-u} \right]. \tag{6.3}$$

Every fixed mixed parameter derivative may be taken termwise. They have the exact representations

$$B_\alpha(u) = \frac{L(u)}{u} - \zeta(1+u, \alpha), \tag{6.4}$$

$$D_\alpha(u) = \frac{L(u)(u+t)}{u(u-1)} - \zeta(u, \alpha) - \kappa_\alpha(u)\zeta(1+u, \alpha). \tag{6.5}$$

Every apparent singularity of these expressions in $\Re u > -1$ is removable. The meromorphic continuation of the two-particle sum is

$$\begin{aligned} Z(u) &= A(u)\,[\zeta(u,\alpha) + \kappa_\alpha(u)\zeta(1+u,\alpha) + D_\alpha(u)] - [\zeta(1+u,\alpha) + B_\alpha(u)]^2 \\ &= -\frac{L(u)^2(1+u/a)(1+u/c)}{u^2(1-u)}. \end{aligned} \tag{6.6}$$

In particular, the square of the subtracted harmonic moment is an essential overlap term.

Proof. The uniform Gamma-ratio expansion on compact parameter sets [10, Section 5.11(iii)] gives

$$w_n(u) = n^{-2-u}\left[1 + \frac{\beta(u)}{n} + O(n^{-2})\right], \qquad \beta(u) = -t - (a+c+\tfrac{1}{2})u - \tfrac{1}{2}u^2.$$

When n is replaced by $n+\alpha$ in the leading power of $n^2 w_n$, the next coefficient becomes $\beta(u)+\alpha u = \kappa_\alpha(u)$. Consequently each summand in (6.2) and (6.3) is $O(n^{-2-\Re u})$, locally uniformly; the same estimate without subtraction applies to (6.1). On a compact subset of $\Re u > -1$ these estimates have a summable common majorant. Cauchy's integral formula on slightly larger compact neighborhoods then gives the assertion for all fixed mixed derivatives. In particular, differentiating at $u = 0$ is justified before any meromorphic expression is evaluated.

For the region in which the relevant ordinary moments converge, Gauss summation [10, Equation 15.4.20] yields

$$\begin{aligned} \sum_{n\ge 0} w_n(u) &= A(u), \\ \sum_{n\ge 0} n w_n(u) &= \frac{L(u)}{u}, \\ \sum_{n\ge 0} n(n-1) w_n(u) &= \frac{(a+1)(c+1)L(u)}{u(u-1)}. \end{aligned}$$

These formulas follow either by shifting the index in the defining ${}_2F_1$ series, or by differentiating it once and twice in its argument before setting that argument to one. Adding the last two moments gives $L(u)(u+t)/(u(u-1))$. Subtracting the two Hurwitz model series proves (6.4) and (6.5) first on nonempty open convergence domains and then by analytic continuation. The proved holomorphy of the left sides proves removability at both $u = 0$ and $u = 1$ wherever they occur in the displayed right sides.

Finite Cauchy–Binet gives $\sum_{0\le m<n\le N}(n-m)^2 w_m w_n = M_{0,N}M_{2,N} - M_{1,N}^2$. Pass to the limit for $\Re u > 1$ and insert the preceding moment formulas. The numerator simplifies using $u(u+t) - ac(u-1) = (u+a)(u+c)$, proving (6.6).

There is also a fully summable two-index remainder, rather than only a formal determinant of continued quantities. Write $b_n(u)$ and $d_n(u)$ for the summands in (6.2) and (6.3). Then

$$A(u)D_\alpha(u) - B_\alpha(u)^2 = \sum_{m,n\ge 0} [w_m(u)d_n(u) - b_m(u)b_n(u)]$$

is absolutely and locally normally convergent, by the product of the three ℓ^1 majorants just proved. The remaining terms in (6.6) are the explicitly displayed model sums and their cross terms. Thus the complete subtraction, including overlaps, has a normally convergent realization independent of expanding the Gamma product. □

Theorem 6.2 (Hard cutoff and spectral finite part)**.** *Let*

$$S_N = \sum_{0\le m<n\le N} (n-m)^2 w_m(0) w_n(0), \qquad x = N+1.$$

As $N \to \infty$, locally uniformly for $\Re a, \Re c > 0$,

$$\boxed{S_N = Ax - (\log x + B)^2 - h(\log x + B) - 2 - A + O\left(\frac{\log^2(x+1)}{x}\right).} \tag{6.7}$$

The Laurent expansion of the spectral continuation is

$$Z(u) = -\frac{1}{u^2} - \frac{2B+h}{u} - \left[2B^2 + T + 2hB + h + A\right] + O(u). \tag{6.8}$$

If C_{cut} denotes the constant after subtraction of $Ax - \log^2 x - (2B+h)\log x$, and C_{sp} the Laurent constant, then

$$C_{\mathrm{cut}} = -B^2 - hB - 2 - A, \tag{6.9}$$

$$C_{\mathrm{sp}} = 2C_{\mathrm{cut}} + 4 + A - h - T. \tag{6.10}$$

This uses the stated unnormalized spectral deformation $w_n(u)$; neither constant is being assigned to an unspecified regulator.

Proof. Use the anchor $\alpha = 1$ in Theorem 6.1. The expansion of $L(u)/u$ at zero gives $B_1(0) = B - \gamma$. Since $\zeta(0) = -1/2$ and $\kappa_1'(0) = 1/2 - a - c$, the constant term in (6.5) gives

$$D_1(0) = -t(B-\gamma) - ac - 1.$$

Consequently the three finite moments satisfy

$$\begin{aligned} M_{0,N} &= A - \frac{1}{x} + O(x^{-2}), \\ M_{1,N} &= \log x + B + O(x^{-1}), \\ M_{2,N} &= x - t\log x - tB - ac - 1 + O(x^{-1}). \end{aligned}$$

The first formula follows by summing the n^{-2} Gamma-ratio tail; the other two follow from the proved summable residuals and $H_x = \log x + \gamma + O(x^{-1})$. Multiplication in the cutoff determinant gives (6.7). In particular, the product of the $-1/x$ correction in $M_{0,N}$ with the x divergence in $M_{2,N}$ contributes -1 to the constant. Omitting it would give an incorrect finite part.

Finally $L(u)^2 = 1 + 2Bu + (2B^2 + T)u^2 + O(u^3)$ and

$$\frac{(1+u/a)(1+u/c)}{1-u} = 1 + hu + (h + A)u^2 + O(u^3).$$

Insert these into (6.6); this proves (6.8) and the comparison formula. □

Dependence on the spectral normalization. Multiplying each particle's weight by a holomorphic function $\lambda(u)$ with $\lambda(0) = 1$ leaves the hard cutoff at zero unchanged. Write $\lambda_j = \lambda^{(j)}(0)$. Since the total spectral sum is multiplied by $\lambda(u)^2$, its residue and constant become, respectively,

$$-(2B+h) - 2\lambda_1, \qquad C_{\mathrm{sp}} - 2\lambda_1(2B+h) - \lambda_1^2 - \lambda_2.$$

Thus the second derivative of the normalization enters at this double pole. These expressions follow by multiplying the two Taylor–Laurent series; they explain exactly why the first-resonance rule, which has only one amplitude derivative, cannot be copied unchanged to the present resonance.

A convergent renormalized sum. Let $K_N = S_N - S_{N-1}$, $S_{-1} = 0$, and $d = B - \gamma$. The following is an ordinary absolutely convergent identity:

$$\sum_{m\geq 1}\left[K_{m-1} - A + \frac{2H_m + 2d + h}{m} - \frac{1}{m^2}\right] = -d^2 - hd - 2 - A. \tag{6.11}$$

Indeed the largest-index formula $K_N = w_N(0)[N^2 M_{0,N-1} - 2NM_{1,N-1} + M_{2,N-1}]$ and one further Gamma-ratio term give a remainder $O(\log(m+1)/m^2)$ in the displayed summand. Its partial sum is exactly $S_{m-1} - Am + H_m^2 + (2d+h)H_m$, and (6.7) gives the limit. The $m = 1$ term is included; the two-particle mass there is zero.

6.1 Uniform spectral deformation and every cutoff jet

The previous calculation extends to a uniform formula which retains the two different escape powers. In particular, derivatives at the resonance can be computed without differentiating a nonuniform asymptotic assertion.

Theorem 6.3 (Uniform near-resonance expansion)**.** *Let $0 < \delta < 1/3$, $x = N + 1$, and $S_N(u) = \sum_{0\leq m<n\leq N}(n-m)^2 w_m(u)w_n(u)$. Uniformly for $|u| \leq \delta$ and for a, c on compact subsets of their right half-planes,*

$$\begin{aligned} S_N(u) = Z(u) + \frac{A(u)x^{1-u}}{1-u} \\ + x^{-u}\left[\frac{2L(u)}{u^2} - \frac{A(u)\beta(u)}{u} - \frac{A(u)}{2}\right] - \frac{x^{-2u}}{u^2(1-u^2)} + O\left(x^{-1+2\delta}\log^2(x+1)\right), \end{aligned} \tag{6.12}$$

where $\beta(u) = -t - (a + c + 1/2)u - u^2/2$. The full displayed expression has a removable singularity at $u = 0$. The error is holomorphic in u near this closed disk; every fixed derivative at $u = 0$ tends to zero as $N \to \infty$.

Proof. The uniform Gamma-ratio expansion from Theorem 6.1 and the first two Euler–Maclaurin terms for the corresponding Hurwitz tails [10, Equation 25.11.43] give

$$\begin{aligned} M_{0,N}(u) &= A(u) - \frac{x^{-1-u}}{1+u} + O(x^{-2+\delta}), \\ M_{1,N}(u) &= \frac{L(u) - x^{-u}}{u} + O(x^{-1+\delta}), \\ M_{2,N}(u) &= \frac{L(u)(u+t)}{u(u-1)} + \frac{x^{1-u}}{1-u} - x^{-u}\left[\frac{1}{2} + \frac{\beta(u)}{u}\right] + O(x^{-1+\delta}). \end{aligned}$$

For a direct justification of the analytic tails, subtract n^{-1-u} from $nw_n(u)$ and subtract $n^{-u} + \beta(u)n^{-1-u}$ from $n^2 w_n(u)$ before summing from $n = x$ to infinity. The residuals are uniformly summable for $\Re u > -1$. Apply the ordinary Hurwitz tail expansion to the model powers, first in their convergence regions and then by analytic continuation. This proves both the error bounds and their uniform removability at $u = 0$.

The apparent quotients involving u^{-1} in the second and third lines, after their pole cancellations, are $O(x^\delta \log(x+1))$. On multiplying the finite moment determinant, the tail of $M_{0,N}$ times the leading divergence of $M_{2,N}$ contributes $-x^{-2u}/((1+u)(1-u))$. All other terms involving a remainder or this tail are $O(x^{-1+2\delta}\log^2(x+1))$. Combine this contribution with the $-x^{-2u}/u^2$ from the square of $M_{1,N}$; their sum is the last term in (6.12). The other terms simplify to the first three displayed

terms by (6.6). Finally, use Cauchy's estimate on a circle of radius δ around zero to transfer the tending-to-zero error to every fixed derivative. For clarity, the possible negative Laurent powers in the full model cancel as follows: their u^{-2} coefficients are $-1+2-1=0$, and their u^{-1} coefficients are $-(2B+h)+(2B+h-2\log x)+2\log x=0$. All other potential poles of the model lie outside $|u|<1$. Thus the difference between the finite cutoff and the model is holomorphic throughout $|u|<1$, as required for Cauchy's estimate. □

In particular, for each integer $q\ge 0$ there are explicit polynomials P_q, Q_q of degrees at most $q, q+2$ such that

$$\partial_u^q S_N(u)|_{u=0} = xP_q(\log x) + Q_q(\log x) + o(1). \tag{6.13}$$

They are obtained by taking $q!$ times the coefficient of u^q in (6.12); only finitely many Gamma derivatives are needed at each q. The highest coefficient of the second polynomial is universal:

$$[y^{q+2}]Q_q(y) = \frac{(-1)^q(2-2^{q+2})}{(q+1)(q+2)}. \tag{6.14}$$

Indeed the only contributions of this degree come from $2e^{-uy}/u^2 - e^{-2uy}/u^2$. This formula includes the $-\log^2 x$ at $q=0$ and the $+\log^3 x$ at $q=1$. The scale x^{-2u} is essential even when only the finite part of the undeformed cutoff is wanted.

6.2 Generalized Stieltjes jets of the subtracted moments

The normally convergent formula has an auxiliary Hurwitz anchor, which supplies genuine generalized Stieltjes identities. This is a consequence of the subtraction theorem, not a claim that merely differentiating Gauss summation is new.

Let Y_q be the exponential complete Bell polynomial, with $Y_0=1$, and set

$$\ell_j = \psi^{(j-1)}(1) - \psi^{(j-1)}(a+1) - \psi^{(j-1)}(c+1), \qquad j\ge 1.$$

For $q\ge 0$, write

$$W_q(n) = w_n(0)Y_q(-\psi(n+b_0), -\psi'(n+b_0), \dots, -\psi^{(q-1)}(n+b_0)).$$

The interpretation is $W_0(n)=w_n(0)$. Then

$$\boxed{\sum_{n\ge 0}\left[nW_q(n) - \frac{(-\log(n+\alpha))^q}{n+\alpha}\right] = \frac{Y_{q+1}(\ell_1,\dots,\ell_{q+1})}{q+1} - (-1)^q\gamma_q(\alpha).} \tag{6.15}$$

All logarithms have the right-half-plane branch. This is ordinary absolute convergence, including every fixed mixed derivative in a, c, α. For example,

$$\sum_{n\ge 0}\left[nw_n(0) - \frac{1}{n+\alpha}\right] = B + \psi(\alpha),$$

$$\sum_{n\ge 0}\left[-nw_n(0)\psi(n+b_0) + \frac{\log(n+\alpha)}{n+\alpha}\right] = \tfrac{1}{2}(B^2+T) + \gamma_1(\alpha).$$

These follow from (6.4) and the defining expansion $\zeta(1+u,\alpha)=u^{-1}+\sum_{q\ge0}(-1)^q\gamma_q(\alpha)u^q/q!$. For the second subtracted moment one may use the similarly explicit coefficient formula

$$\begin{aligned}\frac{D_\alpha^{(q)}(0)}{q!} = & -tp_{q+1}-(t+1)\sum_{j=0}^{q}p_j-\frac{\partial_s^q\zeta(0,\alpha)}{q!}\\ & -\kappa_{q+1}-\sum_{j=0}^{\min(2,q)}\kappa_j\frac{(-1)^{q-j}\gamma_{q-j}(\alpha)}{(q-j)!},\end{aligned}\tag{6.16}$$

where $p_j=Y_j(\ell_1,\dots,\ell_j)/j!$, $\kappa_0=-t$, $\kappa_1=\alpha-a-c-1/2$, $\kappa_2=-1/2$, and $\kappa_j=0$ for $j>2$. The left side is obtained by differentiating the summands of (6.3); thus (6.16) is a finite compiler for convergent Gamma/polygamma sums in generalized Stieltjes constants and ordinary spectral Hurwitz-zeta derivatives.

6.3 A rational specialization and its complete polylogarithm generator

At $a=c=1$, the weight is $w_n(0)=1/((n+1)(n+2))$. With $x=N+1$ the elementary finite moments are

$$M_{0,N}=\frac{x}{x+1},\quad M_{1,N}=H_x+\frac{2}{x+1}-2,\quad M_{2,N}=x-3H_x-\frac{4}{x+1}+4.$$

Their determinant simplifies to

$$\boxed{S_N=x-H_x^2+H_x-1-\frac{H_x-1}{x+1},\qquad K_N=1-\frac{H_x}{x}-\frac{H_x-1}{x+1}.}\tag{6.17}$$

Thus the cutoff has harmonic degree two, but its largest-index generator has depth two. The finite difference removes the constant coefficient of H_x^2; a formal harmonic-degree bound at the cutoff level need not be sharp for the generator.

For $|z|<1$, summation of (6.17) gives

$$\boxed{\sum_{0\le m<n}\frac{(n-m)^2z^{n+1}}{(m+1)(m+2)(n+1)(n+2)}=\frac{z}{1-z}-1-\mathrm{Li}_2(z)-\frac{1+z}{2z}\log^2(1-z)-\frac{\log(1-z)}{z}.}\tag{6.18}$$

The singularities at $z=0$ cancel, and its first term is $z^2/12$. To check the summation, use

$$\sum_{m\ge1}\frac{H_mz^m}{m}=\mathrm{Li}_2(z)+\tfrac12\log^2(1-z),\qquad \sum_{m\ge1}\frac{H_mz^m}{m+1}=\frac{\log^2(1-z)}{2z}.$$

Both follow by integrating the elementary generating function $\sum H_mz^m=-\log(1-z)/(1-z)$. In particular,

$$\sum_{0\le m<n}\frac{(n-m)^2\,2^{-n-1}}{(m+1)(m+2)(n+1)(n+2)}=2\log2-\log^2 2-\frac{\pi^2}{12}.\tag{6.19}$$

For any integer $q\ge2$, let

$$J_q=\sum_{0\le m<n}\frac{(n-m)^2}{(m+1)(m+2)(n+1)^{q+1}(n+2)}.$$

It converges absolutely since $K_N \to 1$. We use the strict convention $\zeta(j,1) = \sum_{m>k\geq 1} m^{-j}k^{-1}$. Put $E_j = \zeta(j,1) + \zeta(j+1)$ for $j \geq 2$, $R_1 = \zeta(2) - 1$, and $R_q = E_q - \zeta(q) - R_{q-1}$ for $q \geq 2$. Then

$$J_q = \zeta(q) - E_{q+1} - R_q. \tag{6.20}$$

Writing $G(z)$ for (6.18) and $\Theta = z\,d/dz$, the corresponding anchored primitives on $|z| < 1$ are

$$\Theta^{-q}G(z) = \sum_{m\geq 1} \frac{K_{m-1}z^m}{m^q} = \frac{1}{(q-1)!}\int_0^z \log^{q-1}(z/t)G(t)\frac{dt}{t} \quad (q \geq 1),$$

initially on a positive real segment and then by the analytic power series. Their endpoints at one are J_q for $q \geq 2$. Indeed $R_q = \sum_{m\geq 1}(H_m - 1)/(m^q(m+1))$, and partial fractions prove its displayed recurrence; the initial value follows by telescoping. Euler's classical reduction of $\zeta(j,1)$, proved in the incoming report [5, Section 6], gives a terminating formula using only ordinary zeta values and products of two such values, with weight at most $q+2$. Explicitly, for $j \geq 2$ that reduction is

$$\zeta(j,1) = \frac{j}{2}\zeta(j+1) - \frac{1}{2}\sum_{r=1}^{j-2}\zeta(j-r)\zeta(r+1),$$

with the empty sum interpreted as zero. Hence (6.20) is a finite ordinary-zeta evaluation at every integer $q \geq 2$. The first two evaluations are

$$\boxed{J_2 = 3\zeta(2) - 2\zeta(3) - \tfrac{5}{4}\zeta(4) - 1,} \tag{6.21}$$

$$J_3 = 1 - 2\zeta(2) + 4\zeta(3) - \tfrac{5}{4}\zeta(4) - 3\zeta(5) + \zeta(2)\zeta(3). \tag{6.22}$$

The primitive $q = 1$ diverges logarithmically and is deliberately excluded from this endpoint statement.

Next questions. The next unsolved step after this result is an arbitrary-rank subtraction at $b - a - c = 2r - 3$, where a two-by-two divergent moment block interacts with a nontrivial lower determinant. A second target is the full family at $b - a - c = 2r - 2 - M$: separate power and logarithmic counterterms before comparing their cutoff constants with spectral Laurent coefficients. The example above also asks for a depth invariant which respects finite differencing, rather than only the degree of a formal cutoff polynomial.

7 Canonical spatial kernels for every spectral coefficient

Let $P \in \mathbb{C}[u_0, \ldots, u_R]$ be fixed and write $f_P(x) = P(\psi(x), \psi'(x), \ldots, \psi^{(R)}(x))$. Its scalar projection is $\phi_P(Y) = P(Y, 0, \ldots, 0)$. For a sufficiently large finite M, define the principal-part data

$$p_j(n) = [v^{-j}]f_P(-n+v), \qquad P_0(x) = \sum_{j=1}^{M} p_j(0)x^{-j}, \qquad \Delta p_j(n) = p_j(n) - p_j(n-1).$$

All $p_j(n)$ are explicit polynomials in ordinary harmonic numbers and zeta values. Define the harmonic Dirichlet functions by

$$D_j(s) = \sum_{n\geq 1} \Delta p_j(n) n^{-s} \qquad (\Re s > 0). \tag{7.1}$$

The subscript j denotes pole order, not differentiation order. We are interested in every Laurent coefficient $[s^q]D_1(s)$, $q \geq 1$. The previously established unweighted finite-part bridge uses only $q = 1$ [6, 8].

The normalized spatial function and its leading tail are

$$h_P(x) = f_P(x+1) - f_P(x) + P_0(x), \qquad K_P(L) = \phi_P'(L) + p_1(0) = \sum_{k\geq 0} K_k L^k. \tag{7.2}$$

The P_0 term removes the entire singularity at $x = 0$. In particular $h_P(0) = f_P(1) - [x^0]f_P(x)$.

Theorem 7.1 (Normalized Mellin identity)**.** *The integrals*

$$E_P(s) = \int_0^1 x^{-s} h_P(x)\,dx + \int_1^\infty x^{-s}\left\{h_P(x) - \frac{K_P(\log x)}{x}\right\} dx \tag{7.3}$$

define a holomorphic function on $-1 < \Re s < 1$. On that strip, as an identity of meromorphic functions,

$$\boxed{D_1(s) = -\frac{\sin \pi s}{\pi}\left(E_P(s) + \sum_{k\geq 0}\frac{K_k k!}{s^{k+1}}\right) - \sum_{j=2}^{M}\frac{(s)_{j-1}}{(j-1)!}D_j(j-1+s).} \tag{7.4}$$

The rising factorials are finite polynomials. Every shifted D_j in the second sum is holomorphic at $s = 0$, with all its derivatives represented by absolutely convergent series.

Proof. The inherited normalized principal-part expansion [6, Section 2] gives, on $x > 0$, the particularly simple shift identity

$$h_P(x) = -\sum_{n\geq 1}\sum_{j=1}^{M}\frac{\Delta p_j(n)}{(n+x)^j}. \tag{7.5}$$

It can also be obtained directly by shifting a finite cutoff of the principal-part expansion: the last boundary term tends to zero since $p_j(n) = O((1+\log n)^d)$. The local Laurent expansion proves analyticity at $x = 0$. The differentiable positive-axis polygamma asymptotic expansions give

$$f_P(x+1) - f_P(x) = \frac{\phi_P'(\log x)}{x} + O((1+\log x)^d/x^2),$$

and hence $h_P(x) = K_P(\log x)/x + O((1+\log x)^d/x^2)$. These estimates prove normal convergence of (7.3), together with every fixed s-derivative, on its stated strip.

For $0 < \Re s < 1$, integrate (7.5) against x^{-s}. The exact beta integral is

$$\int_0^\infty \frac{x^{-s}}{(n+x)^j}\,dx = n^{1-s-j}\frac{\Gamma(1-s)\Gamma(j-1+s)}{\Gamma(j)} = \frac{\pi}{\sin \pi s}\frac{(s)_{j-1}}{(j-1)!}n^{-(j-1+s)}.$$

Fubini is justified by $\Delta p_j(n) = O((1+\log n)^d/n)$; the slowest case, $j = 1$, is summable for $\Re s > 0$ after taking the beta integral. Also

$$\int_1^\infty x^{-s-1}K_P(\log x)\,dx = \sum_{k\geq 0} K_k k! s^{-k-1}.$$

Solving for $D_1(s)$ gives (7.4) on $0 < \Re s < 1$, and its right side supplies continuation to the whole strip. For $j \geq 2$, $j - 1 + s$ has positive real part throughout $\Re s > -1$, so its series is holomorphic there. This also proves the assertions about ordinary derivatives. □

For $q \geq 1$, define the universal polynomial

$$k_q(L) = \frac{(-1)^q}{q!}\frac{(L+i\pi)^q - (L-i\pi)^q}{2i\pi} = (-1)^q \sum_{m=0}^{\lfloor (q-1)/2 \rfloor} \frac{(-1)^m \pi^{2m} L^{q-2m-1}}{(2m+1)!(q-2m-1)!}. \tag{7.6}$$

The first four are

$$k_1 = -1, \qquad k_2 = L, \qquad k_3 = \frac{\pi^2}{6} - \frac{L^2}{2}, \qquad k_4 = \frac{L^3 - \pi^2 L}{6}.$$

For any polynomial w, put

$$\mathcal{L}_P[w] = \int_0^1 w(\log x) h_P(x)\, dx + \int_1^\infty w(\log x) \left\{ h_P(x) - \frac{K_P(\log x)}{x} \right\} dx.$$

Both integrals are ordinary absolutely convergent integrals. Set

$$A_{P,q} = - \sum_{\substack{k \geq 0 \\ q+k \text{ even}}} \frac{(-1)^{(q+k)/2} \pi^{q+k} K_k k!}{(q+k+1)!},$$

$$T_{P,q} = \sum_{j=2}^{M} \sum_{n \geq 1} \frac{\Delta p_j(n)}{n^{j-1}} [s^q] \left\{ \frac{(s)_{j-1}}{(j-1)!} e^{-s \log n} \right\}. \tag{7.7}$$

The inner coefficient is an explicit polynomial in $\log n$, so $T_{P,q}$ is an absolutely convergent spectral sum. Equivalently it is a finite rational combination of the derivatives $D_j^{(r)}(j-1)$, $0 \leq r \leq q-1$.

Corollary 7.2 (Every Laurent coefficient has a normalized spatial kernel)**.** *For every fixed differential polynomial P and every integer $q \geq 1$,*

$$\boxed{[s^q] D_1(s) = \mathcal{L}_P[k_q] + A_{P,q} - T_{P,q}.} \tag{7.8}$$

All terms on the right are fixed by ordinary integrals, convergent series, and finite endpoint/polynomial calculations.

Proof. Expand $\sin(\pi s)/\pi$ in (7.4) and differentiate the normally convergent integrals coefficientwise. The coefficient of $-\sin(\pi s) x^{-s}/\pi$ is exactly $k_q(\log x)$. Multiplication with $K_k k!/s^{k+1}$ gives $A_{P,q}$ with the stated sign and parity. The remaining finite sum gives $T_{P,q}$. □

7.1 The same scalar projection controls every Stieltjes correction

Define

$$R_P(L) = -\frac{\phi_P(L+i\pi) - \phi_P(L-i\pi)}{2i\pi}, \qquad R'_P(L) = \sum_{r \geq 0} a_r L^r.$$

The inherited scalar-residue asymptotic [6, Section 2] is

$$\Delta p_1(n) = \frac{R'_P(\log n)}{n} + O((1+\log n)^d / n^2).$$

With $\zeta(1+s) = s^{-1} + \sum_{m \geq 0} (-1)^m \gamma_m s^m / m!$, the following all-order answer follows.

Corollary 7.3 (Complete Stieltjes correction)**.** *For $q \geq 1$,*

$$\boxed{[s^q]D_1(s) = \frac{(-1)^q}{q!}\left\{\sum_{r\geq 0} a_r\gamma_{q+r} + \sum_{n\geq 1}\left(\Delta p_1(n) - \frac{R_P'(\log n)}{n}\right)(\log n)^q\right\}.} \tag{7.9}$$

The second sum converges absolutely. In particular a mixed polynomial, for which $\phi_P = 0$, requires no Stieltjes correction at any order.

Proof. Subtract the leading polynomial from the Dirichlet series:

$$D_1(s) = \sum_r a_r(-1)^r\zeta^{(r)}(1+s) + \sum_{n\geq 1}\left(\Delta p_1(n) - R_P'(\log n)/n\right)n^{-s}.$$

The second series is holomorphic for $\Re s > -1$. The coefficient of s^q in $(-1)^r\zeta^{(r)}(1+s)$ is $(-1)^q\gamma_{q+r}/q!$. This gives the formula. □

Recovery of the original finite-part normalization. For $q = 1$, $T_{P,1} = \sum_{j\geq 2,n\geq 1}\Delta p_j(n)/((j-1)n^{j-1}) = J_P$. If $C_P = (2\pi)^{-1}\int_{-\pi}^{\pi}\phi_P(i\theta)\,d\theta$, then direct coefficient extraction gives $A_{P,1} = \phi_P(0) - C_P$. Thus the inherited identity $\mathrm{FP}\int_0^1 f_P = C_P + J_P + [s]D_1(s)$ becomes

$$\mathrm{FP}\int_0^1 f_P(x)\,dx = \phi_P(0) - E_P(0). \tag{7.10}$$

This is a normalization check; no endpoint constant has been dropped. For $q \geq 2$, (7.8) supplies the spatial normalization requested in Differential Section 09. The choice of the splitting point 1 and the full term $K_P(\log x)/x$ is part of the definition. Altering it without transporting the elementary correction would change the reported constants.

7.2 Two explicit logarithmic integral evaluations

For $P = u_0^2$, the pole differences are $\Delta p_1(n) = -2/n$, $\Delta p_2(n) = 0$, hence $D_1(s) = -2\zeta(1+s)$ and $T_{P,q} = 0$. Moreover

$$h_P(x) = \frac{2(\psi(1+x)+\gamma)}{x}, \qquad K_P(L) = 2L + 2\gamma.$$

The case $q = 2$ gives the ordinary Stieltjes identity

$$\boxed{\int_0^1 \frac{\log x\,[\psi(1+x)+\gamma]}{x}\,dx + \int_1^\infty \frac{\log x\,[\psi(1+x)-\log x]}{x}\,dx = -\frac{\gamma_2}{2} - \frac{\pi^2\gamma}{6}.} \tag{7.11}$$

The higher q versions follow from the printed universal kernels. These special-function ingredients have classical antecedents; the claim here is the all-polynomial normalization theorem, not historical priority for this isolated scalar integral.

For $P = u_1^2$, the principal parts are

$$p_4(n) = 1, \quad p_2(n) = 2(H_n^{(2)} + \zeta(2)), \quad p_1(n) = 4(H_n^{(3)} - \zeta(3)), \quad p_3(n) = 0.$$

Thus $D_1(s) = 4\zeta(3+s)$, $D_2(s) = 2\zeta(2+s)$, $T_{P,2} = 2\zeta'(3)$, and $K_P = -4\zeta(3)$. Write

$$h(x) = -\frac{2}{x^2}(\psi'(1+x) - \zeta(2) + 2\zeta(3)x).$$

The bracket has a double zero at zero. Equation (7.8) gives

$$\boxed{\int_0^1 (\log x)h(x)\,dx + 2\int_1^\infty \frac{\log x}{x^2}(\zeta(2) - \psi'(1+x))\,dx = 2\zeta''(3) + 2\zeta'(3) + \frac{2\pi^2}{3}\zeta(3).} \tag{7.12}$$

Both displayed integrals converge absolutely. This example explicitly tests the origin subtraction, the higher-pole term $T_{P,2}$, and the π^2 endpoint correction independently of the scalar Stieltjes case.

8 Source audit, integration decisions, and further research

8.1 The inspected baseline and the scope of the audit

The comparison baseline is the repository revision

`446dca9bd65c20e409d1d2561d4ffad408103e2a`.

The two locations named in the research request are the canonical polylogarithm manuscript and the current incoming directory [1, 2]. The six current incoming archives were extracted and their relevant statements compared with the manuscript's overview and proof-status records. The targeted mathematical checks concerned the even-tail coefficient conventions and operator reduction, the differential-polynomial normal form and known Gamma coefficient sectors, the normalized residue calculus, and the first determinantal resonance and its stated continuation problem [6, 8, 3, 5].

The new arguments were checked against those inputs, including the omitted binomial summand, the signs and index shifts in the boundary determinant, the Euler-operator projection, and the endpoint constants in the convergent integral formulas. This is a targeted audit of the proof chain used here. It is not a certification of every chapter, historical archive, or earlier numerical calculation in the repository. **No new false inherited theorem was established by this audit.** The source changes below concern proved extensions, attribution, and precise status updates rather than a newly discovered counterexample.

8.2 What is new relative to that baseline

The following distinctions prevent an already available reduction from being counted as a new resolution. "New" here means established in this continuation relative to the inspected source material; it does not assert historical priority over the entire literature.

Topic	Inherited input	Exact advance in this article
Even-tail rank	Equal and adjacent orders; the $p = 2, 4, 6$ families; the fixed-gap boundary determinant in DPL, `ar:gap-det` [6, 8, 3].	Strict separation of every pair of recurrence-column ratios, and nonvanishing of the complete $q - p = 4$ family. The fixed-gap determinant is rederived as an input, not claimed as new.
Finite telescopers	The Bernoulli coordinate formula and explicit low-order telescopers, including $(4, 6)$ [6].	A polynomial certificate for every late-block combination and every even pair, with a uniform construction, degree bounds, and limiting constant. The $(8, 12)$ rational extraction is an explicit application.
Gamma coefficients	The canonical quotient, Appell transport, the complete u_0, u_1 sector, and every pure polygamma-square coefficient [6, 8, 3].	The complete projection onto u_0, u_1, u_2, including all powers u_2^b, $b \geq 2$, and the rational coefficient form with numerator degree $b - 2$. These are formal coefficient identities.
Next resonance	Classical Gauss summation and the determinant identity; the first-resonance theory and proposed next two-particle target [5].	The next double pole, including overlap subtraction, the hard-cutoff constant, its spectral comparison, uniform cutoff jets, and anchored Hurwitz–Stieltjes identities.
Higher spectral jets	The normalized principal-part expansion, scalar projection, and first spectral-coefficient bridge [6].	Canonical spatial kernels for every $[s^q]D_1(s)$, $q \geq 1$, with all origin, infinity, and higher-pole terms. The same scalar projection controls every Stieltjes correction.
Family–2 example	An exact incoming proof of the manuscript's `double:eq:fam2-example`, with a simplified value [8].	An editorial promotion supported by that inherited proof. No new proof of this example is claimed here.

8.3 Concrete integration decisions and limits

The optimal-prefix conjecture should now record the union of the proved equal-order, adjacent-order, $p = 2, 4, 6$, and gap-four families. Its general unequal-order statement remains open. In particular, pairwise ratio separation does not settle a boundary determinant of size three or larger. There is a simple algebraic reason: the positive matrix $B_{ij} = i + j$, $1 \leq i, j \leq 3$, has every increasing two-by-two minor strictly negative,

$$\det B_{\{i_1,i_2\},\{j_1,j_2\}} = -(i_2 - i_1)(j_2 - j_1) < 0,$$

but its three-by-three determinant is zero. Thus the new ratio theorem supplies exactly the sign information needed for a two-column boundary block; an additional argument is required for $m \geq 3$.

The rational telescoper theorem changes the status of the requested all-order certificate construction. It does not by itself prove that an arbitrary prescribed inverse row exists. When a coefficient relation has been supplied, the theorem constructs and verifies its finite certificate; the separate rank theorem proves existence of every reconstruction row in the stated families.

For the Gamma sector, retain the order of operations: first take the canonical normal form and then apply the projection σ_2 which sets $u_r = 0$ for $r \geq 3$. The Euler-operator proof justifies extracting this projected sector; setting higher derivatives to zero before normal reduction is a different operation. The all-b recurrence and rational numerator theorem extend the earlier sector results. They neither evaluate every associated finite-part moment nor establish independence of the resulting numerical constants [6, 8].

All normalizations should travel with their formulas. The tail identities use the centered coordinate $n + 1/2$ and explicit polynomial subtraction. Differential-polynomial finite parts retain the original spatial coordinate and the endpoint functional of total derivatives. The logarithmic kernels use the displayed subtraction at infinity and the splitting point 1. The resonant hard cutoff uses $x = N + 1$, while its spectral constant belongs to the explicitly stated Gamma deformation. The product of a vanishing tail with a divergent moment contributes to that constant and must be retained. These distinctions are requirements of the proved identities, not interchangeable definitions of one unspecified regularization.

The canonical Family–2 display `chapters/04-depth.tex`, `double:eq:fam2-example`, can be marked proved by citing the existing incoming component evaluations `bip:eq:12`–`bip:eq:21` and simplification `bip:eq:family2-short` [8]. This promotion was already justified before the present work. The Gaussian S_6 and the *current* S_8 candidate remain conjectural. The older rejected S_8 vector is a different statement and must not replace the current candidate in either a proof attempt or a status table [1, 8]. None of the new rank, Gamma-sector, or resonance results proves those period identities.

Exact finite algebra and numerical diagnostics serve different purposes. The rational replays test coefficient transformations and finite certificates. The separately subtracted quadratures and complex-parameter comparisons test analytic formulas and endpoint constants. The infinite-family results rest on the proofs, rather than on the extent of a computed table; the floating-point diagnostics are not interval certificates. No proof-assistant formalization is claimed.

8.4 Twelve concrete questions for the next continuation

The following targets emphasize exact identities, structural reductions, and normalized primitives. Each calls for a proof or a specified finite certificate; a failed numerical relation search would not settle an independence or nonmembership claim.

1. **Settle the next boundary determinant.** For $m = 3$, prove or disprove nonvanishing of $\det(C^{[3]}_{b+i,j})_{1\le i,j\le 3}$ for every $b > 3$. This is the next gap $q - p = 6$. The pairwise signs are now known, but they do not control the three-column determinant. A discrete Wronskian identity, an exact recurrence for the determinant, or a nonvanishing arithmetic normalization would supply the missing step.

2. **Explain the gap-four determinant sequence arithmetically.** Find a generating function, finite recurrence, or exact denominator formula for $\Delta_b = u_{b+2}v_{b+1} - u_{b+1}v_{b+2}$. The recurrence already proves positivity; the new target is a closed identity or an integer normalization which explains the numerators of the optimal-prefix determinants and supports exact reconstruction.

3. **Describe all reconstruction rows without generic inversion.** For the proved gap-four family, express each inverse row through finite binomial or Bernoulli sums and the two fundamental recurrence solutions. Combine it with the universal telescoper to obtain certificates for prescribed even-zeta coordinates. Determine minimal weight-polynomial degrees within the specified formal coefficient system, keeping this separate from arithmetic independence.

4. **Extend polynomial telescoping to three or more tails.** Construct a finite difference certificate involving products of every required proper subset of the tails. Identify precisely which parity conditions leave only even reciprocal powers and which additional Euler-sum coordinates survive. The target is an all-order certificate and its endpoint constant, rather than an unproved extrapolation of the two-tail cancellation.

5. **Determine the complete next Gamma projection.** Apply the Euler-operator method to σ_3, retaining u_0, u_1, u_2, u_3. Derive a closed coefficient equation with the canonical support restrictions, then solve the first coupled families containing u_3^2 and arbitrary powers of u_1, u_2. Prove that discarded higher canonical monomials cannot contribute before extracting this sector.

6. **Find a closed formula for the all-b sector numerators.** The rational form, denominator, degree, and leading coefficient of the polynomials $p_b(k)$ are now established. Seek a finite formula uniform in both b and k, a bivariate generating function, or a hypergeometric representation. Such a formula should explain exact vanishing coefficients and coefficient patterns, rather than merely reproduce the already proved recurrence.

7. **Evaluate the numerical image of the new Gamma sector.** Map the moments containing $(\psi')^{2k}(\psi'')^b$ and their Gamma–Appell transports into convergent free-order multiple-zeta derivatives. Determine the relations among these images within an explicitly stated functional-relation family. This would connect the new formal coefficient theorem with the remaining basis-reduction questions for higher mixed moments in the incoming reports [6, 8].

8. **Introduce rational shifts into the logarithmic kernels.** Replace the unshifted pole lattice by a fixed Hurwitz lattice and derive the exact analogue of the scalar correction for $\gamma_q(a)$. At rational a, seek finite formulas in a declared multiple-Hurwitz or cyclotomic basis. Retain the origin subtraction, the choice of logarithm, and the transport of the normalization constant under changes of base point.

9. **Construct parameter primitives for the higher jets.** Starting with the normally convergent kernels, integrate in a shift or Gamma parameter and anchor the primitive at a specified point. Find recurrences which isolate one prescribed higher Stieltjes coefficient or one marked spectral derivative. A complete answer should prove the endpoint limit and identify any Barnes-type functions needed for the resulting primitive family.

10. **Treat higher-rank and deeper determinantal resonances.** At the next general-rank resonance $b - a - c = 2r - 3$, construct the subtraction for the divergent two-by-two moment block together with the finite lower block. Then study the deeper loci $b - a - c = 2r - 2 - M$. Determine all overlapping escape terms before comparing hard-cutoff constants with spectral Laurent coefficients, and prove uniformity sufficient for every fixed parameter jet.

11. **Find a depth description compatible with finite differences.** For rational Gamma weights at other positive integer parameters, derive complete polylogarithm generators of the largest-index increments and their anchored Euler primitives. Determine which cutoff harmonic products cancel under finite differencing and when the endpoint sums reduce to ordinary zeta products. The two-particle specialization here supplies a test case for a depth invariant that respects this operation.

12. **Return to the frozen Gaussian period targets.** For the exact S_6 vector and current S_8 candidate, seek a parameter-dependent polylogarithm or iterated-integral identity whose proved boundary value is the required relation. Preserve the branch data and all lower-weight constants. A projection outside a selected formal relation space is not a disproof of the numerical period identity, and a small residual is not its proof [1].

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges: Special Values, Zeta Jets, Lattices and Experimental Discovery*. Consolidated manuscript, October 2026. Repository snapshot `446dca9bd65c20e409d1d2561d4ffad408103e2a`. `Analysis/Polylogarithms/docs/manuscript`.

[2] *Incoming reports*. ProveIt repository inventory and intake instructions, at snapshot `446dca9bd65c20e409d1d2561d4ffad408103e2a`. `docs/incoming`; pinned intake README. The six archives below are repository working reports; their presence in the incoming directory does not confer canonical or formal status.

[3] *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals: Exact harmonic-tail reconstruction, polylogarithmic boundary identities, and finite-part reduction through the seventh power*. Repository working report, 11 October 2026 (UTC). `ProveIt_Bernoulli_Tornheim_Digamma_2026-10-11.zip`.

[4] *Cubic Invariant Tail Descent: All-degree harmonic identities, exact digamma cubes, polylogarithmic representations, and algebraic Stieltjes traces*. Repository working report, 10 October 2026. `ProveIt_Cubic_Tail_Descent_2026-10-10.zip`.

[5] *Determinantal Harmonic Resonance: Gauss–Vandermonde finite parts, Stieltjes jets, and depth-two polylogarithmic closure*. Repository working report, 10 October 2026. `ProveIt_Determinantal_Harmonic_Resonance_2026-10-10.zip`.

[6] *Differential Polygamma and Lerch Identities: Exact residue cancellation, higher Gamma moments, spectral derivatives, and harmonic-tail reconstruction*. Repository working report, 11 October 2026. `ProveIt_Differential_Polygamma_and_Lerch_Jets_2026-10-11.zip`.

[7] *Contact-Complete Multishift Lerch Reflection: Independent spectral jets, generalized Stieltjes log moments, confluent contacts, and Gamma–polylogarithm identities*. Repository working report, 10 October 2026. `ProveIt_Multishift_Reflection_2026-10-10.zip`.

[8] *Exact Polylogarithm Parity, Harmonic-Tail Reconstruction, and Higher Polygamma Moments: Conjecture proofs, canonical reductions, spectral derivatives, and a source-level research audit*. Repository working report, 11 October 2026. `ProveIt_Polygamma_Tail_Identities_2026-10-11.zip`.

[9] Li Guo, Georg Regensburger, and Markus Rosenkranz, *On integro-differential algebras*, Journal of Pure and Applied Algebra **218**, no. 3 (2014), 456–473. `https://doi.org/10.1016/j.jpaa.2013.06.015`. Preprint: `https://arxiv.org/abs/1212.0266`.

[10] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*. Version 1.2.8, released 15 September 2026; consulted 11 October 2026. In particular, §5.7 (local Gamma series), §5.11 (Gamma ratios), §5.15 (polygamma functions), §15.4.20 (Gauss summation), §24.8.1 (Bernoulli Fourier series), and §25.11 (Hurwitz zeta). `https://dlmf.nist.gov/5.7`; `https://dlmf.nist.gov/5.11.E13`; `https://dlmf.nist.gov/5.15`; `https://dlmf.nist.gov/15.4.E20`; `https://dlmf.nist.gov/24.8.E1`; `https://dlmf.nist.gov/25.11`.